

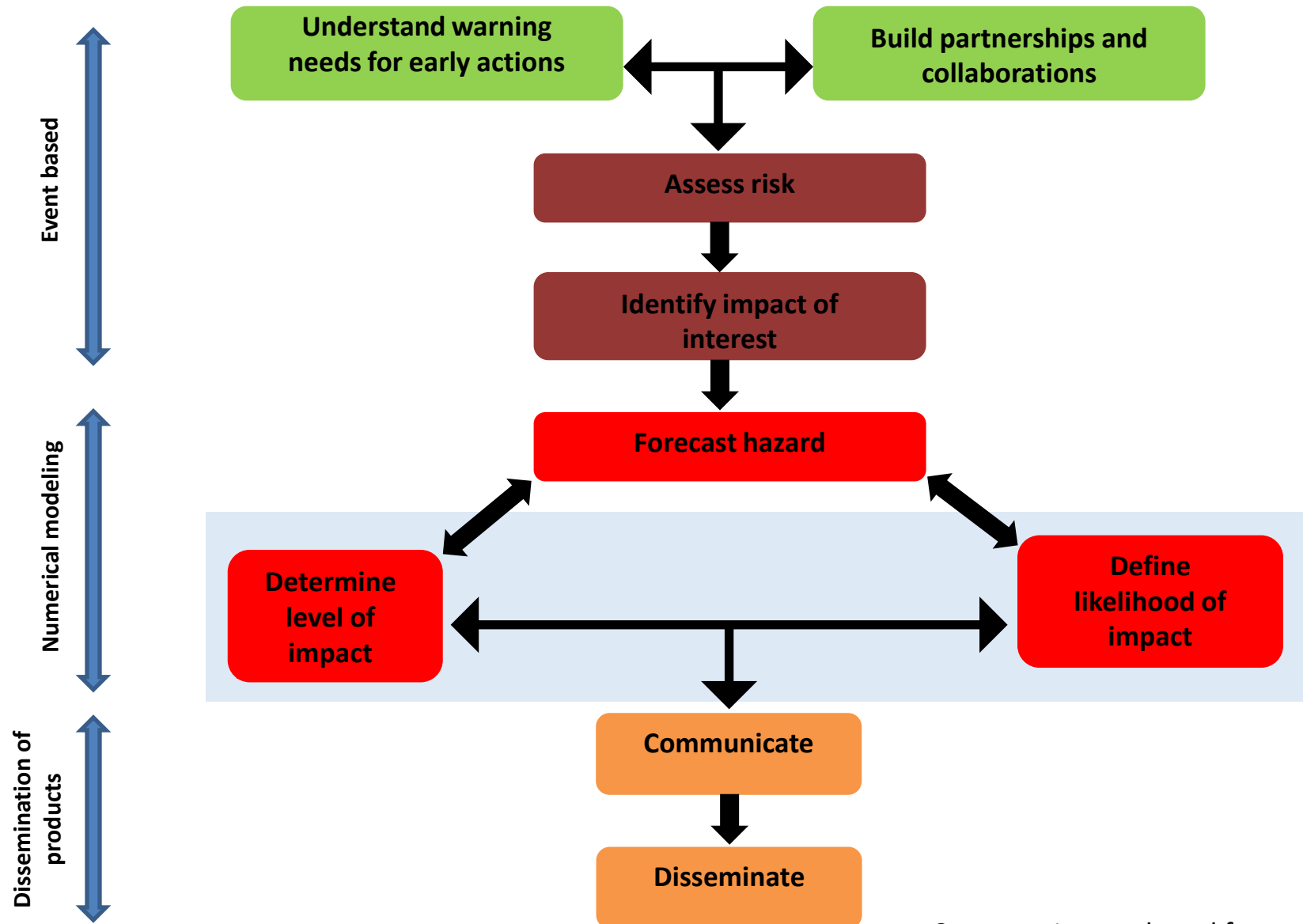
Real-time hazard forecasting from INCOIS



**Dr. PLN Murty, Dr. K Siva Srinivas, Mr. J Padmanabham,
Dr. P G Remya, Dr. S J Prasad**

Indian National Centre for Ocean Information Services (INCOIS)

Impact Based Forecast: Impact-based forecasting provides **the information needed to act before disasters to minimize** the socio- economic costs of weather and climate hazards. The inclusion of risk assessments makes impact-based forecasting unique among other forecasts and warnings.



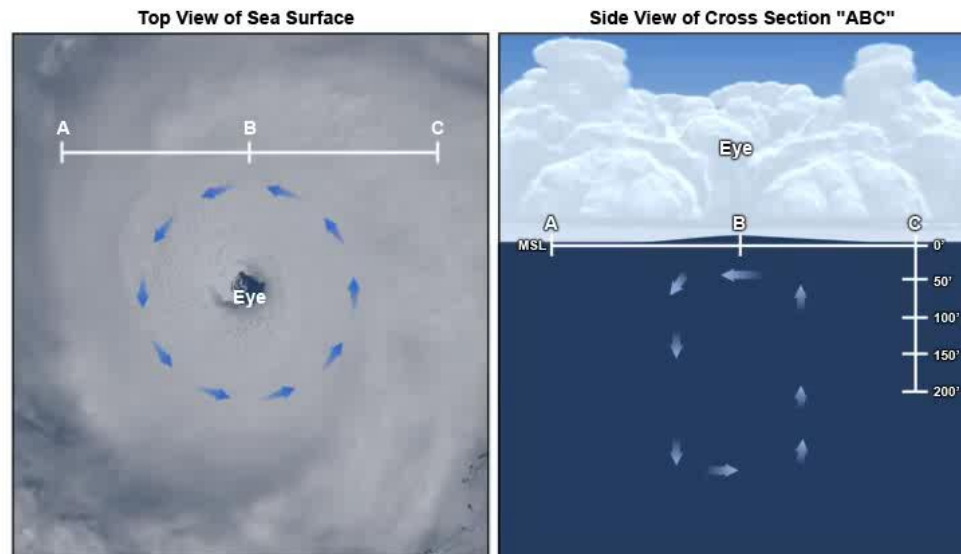
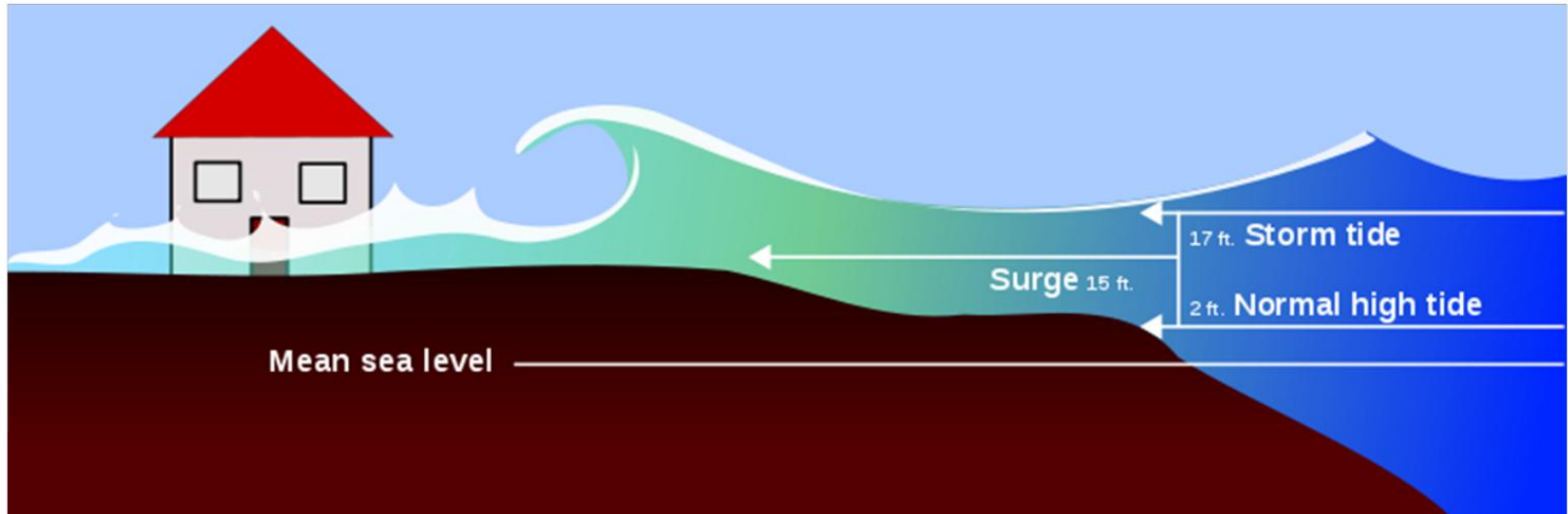
Hazards and the models under use for respective forecasts

Hazard	MODEL	STATUS
Storm surge and associated inundation	ADCIRC	Operational
	ADCIRC+SWAN	Experimental
Tsunami propagation	Tunami	Operational
Tsunami propagation & inundation	ADCIRC	Experimental
Wave	SWAN (Near Shore) WW3 (Global, regional)	Operational
Oil Spill	GNOME	Operational
Swell Surge	ADCIRC+SWAN	Experimental

Storm Surge

Storm surge

Storm surge is an abnormal raise in sea level at the coast due to a high winds of a tropical storm.

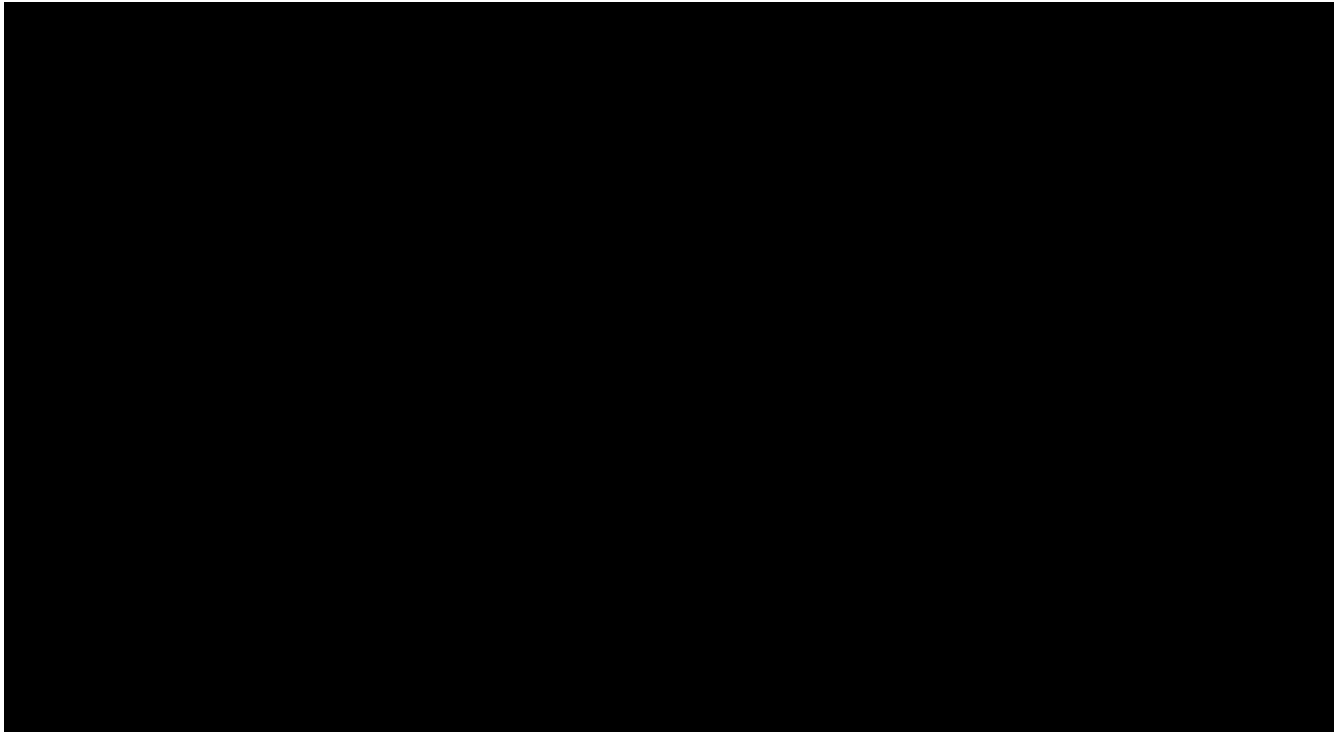


What is coastal inundation?

Coastal inundation is the flooding of normally dry, low-lying coastal land, primarily caused by severe weather events along the coasts, estuaries, and adjoining rivers.

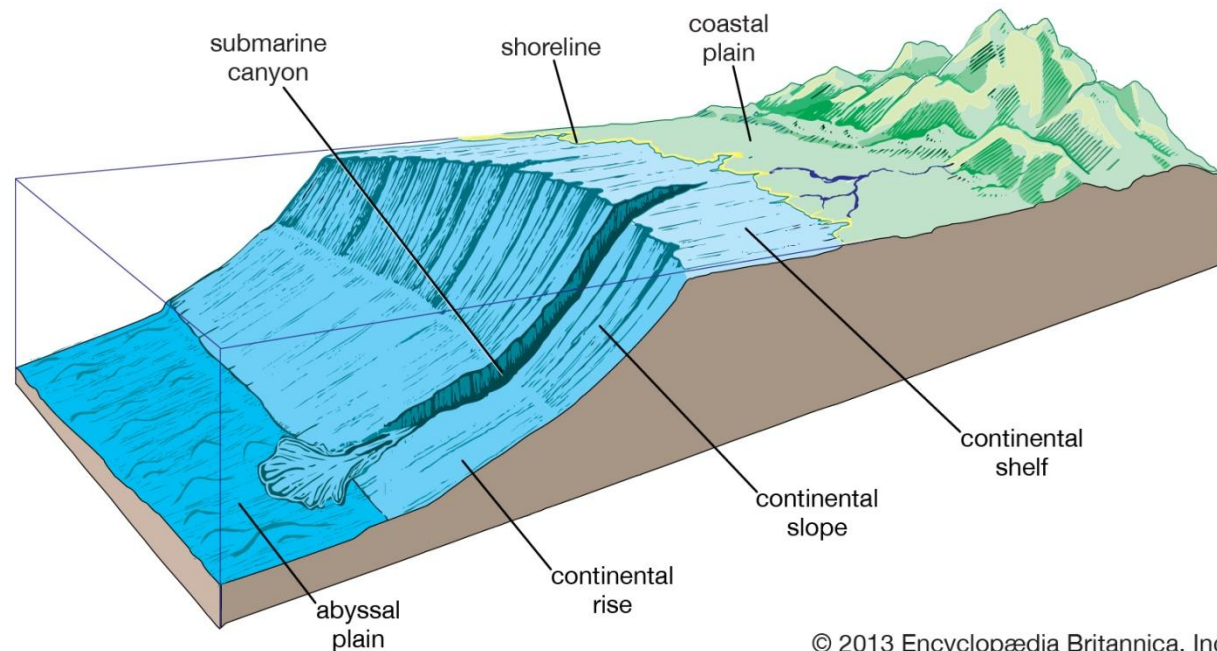
What areas are vulnerable to coastal inundation?

All low-lying coastal regions, which can cover tens of miles inland, are vulnerable to flooding from storms, and the impact can be substantial.

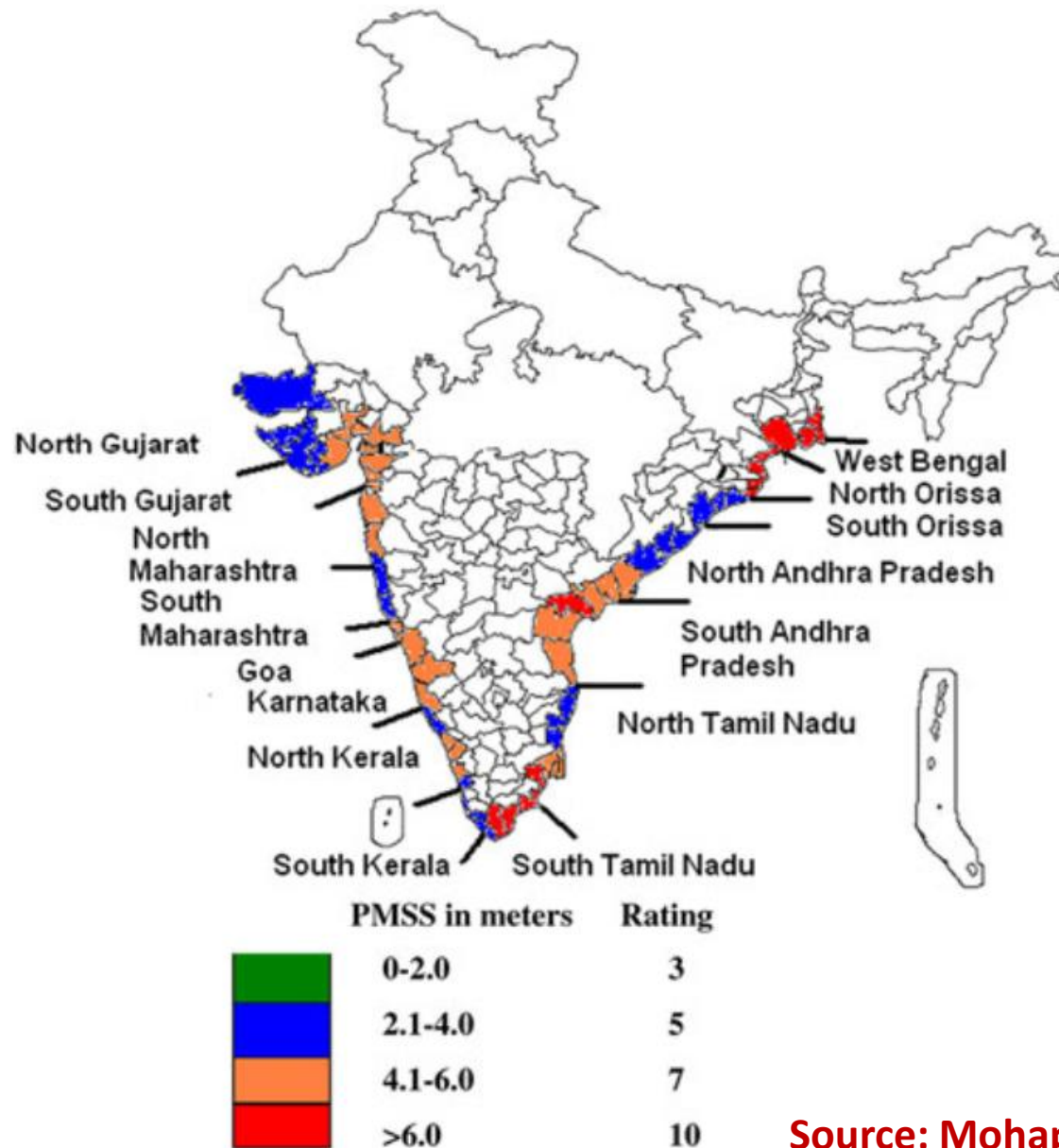


Factors contributing to storm surge:

- Wind
- Storm forward speed
- Low central storm pressure over the ocean
- Tides – phase of the tides contribute to storm surge height
- Slope and width of the continental shelf
 - wide, shallow shelves are prone to larger storm surges.
- Coastal geometry



Probable maximum storm surge over coastal districts of India



Source: Mohapatra et al. 2012

Storm surge model

ADCIRC – ADCIRC is a shallow water hydrodynamic finite element based storm surge model

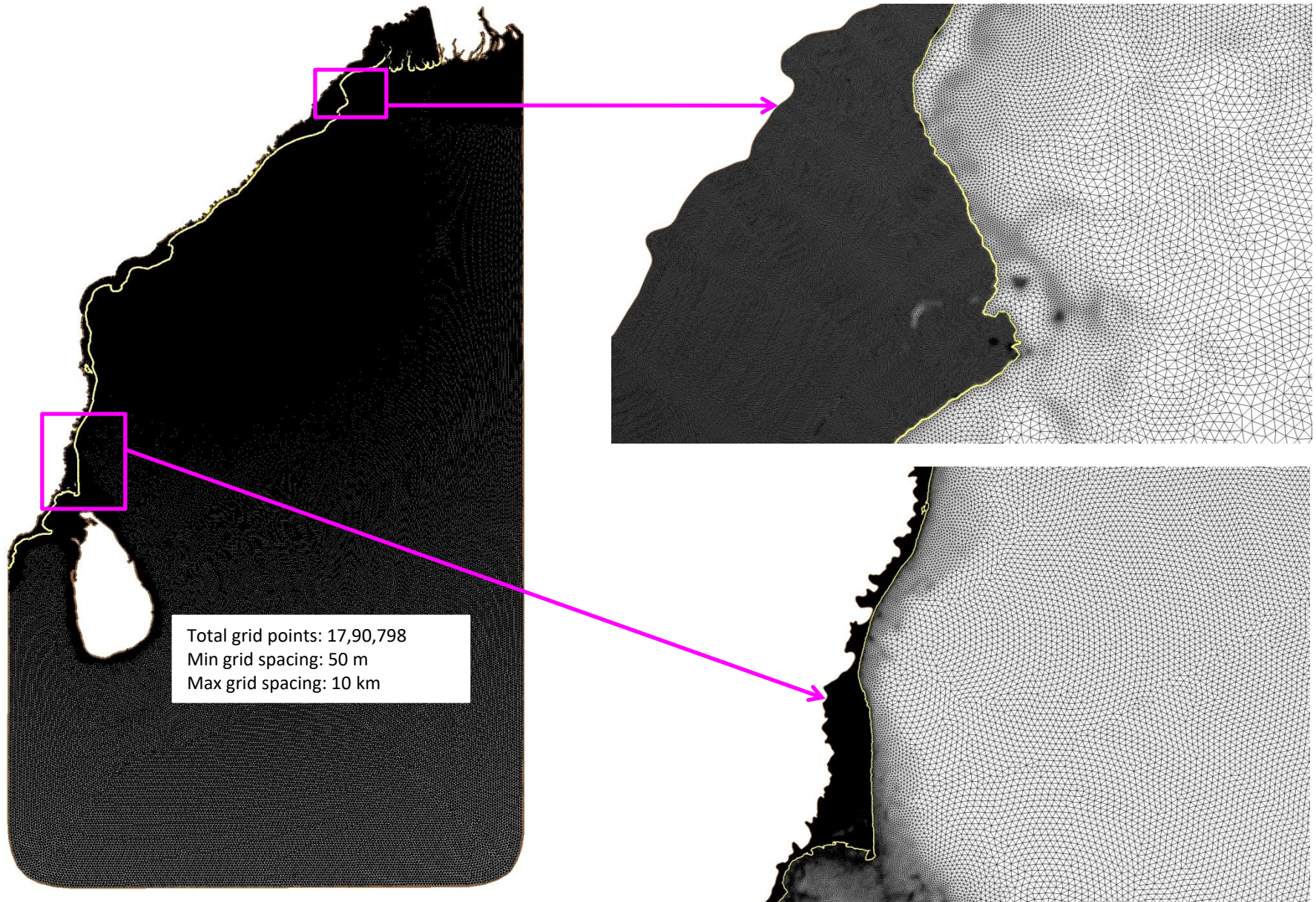
The development of ADCIRC was a joint effort between US Army Corps of Engineers, University of North Carolina and University of Notre Dame (Luettich et al., 1992; Luettich and Westerink, 1991; Westerink and Luettich, 1991; Westerink and Gray, 1991)

ADCIRC applications include modeling tides, seiches and storm surges and their associated inland inundation.

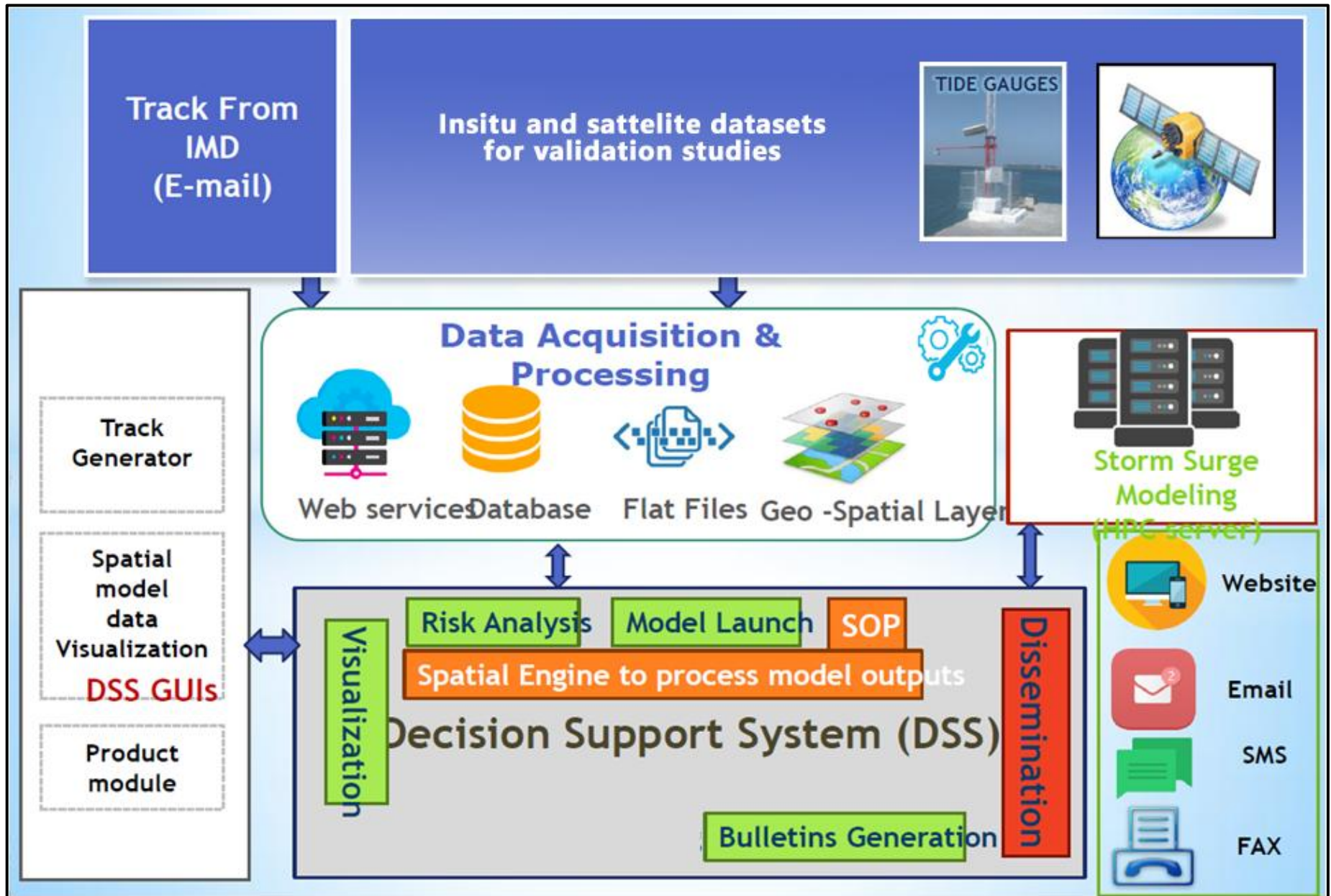
Additional capabilities - wetting and drying algorithm to study inland penetration of water from storm surge

FEMA (Federal Emergency Management Agency) in 2002 accepted the robustness of ADCIRC

Model mesh using for the real-time forecasts (east coast set-up)



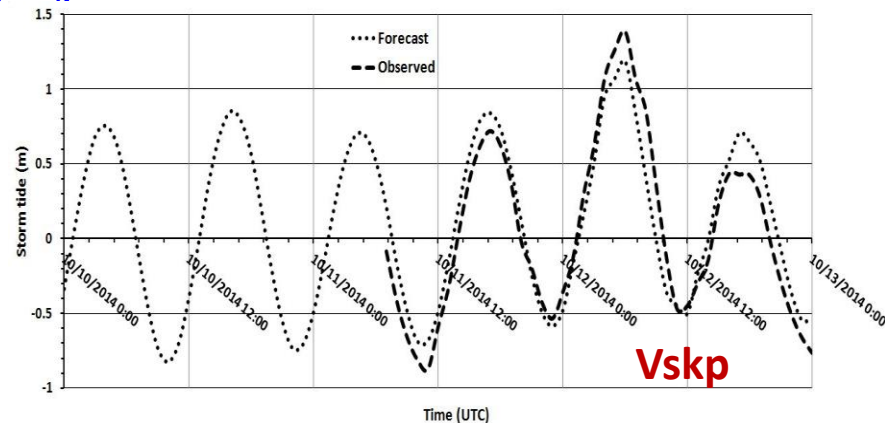
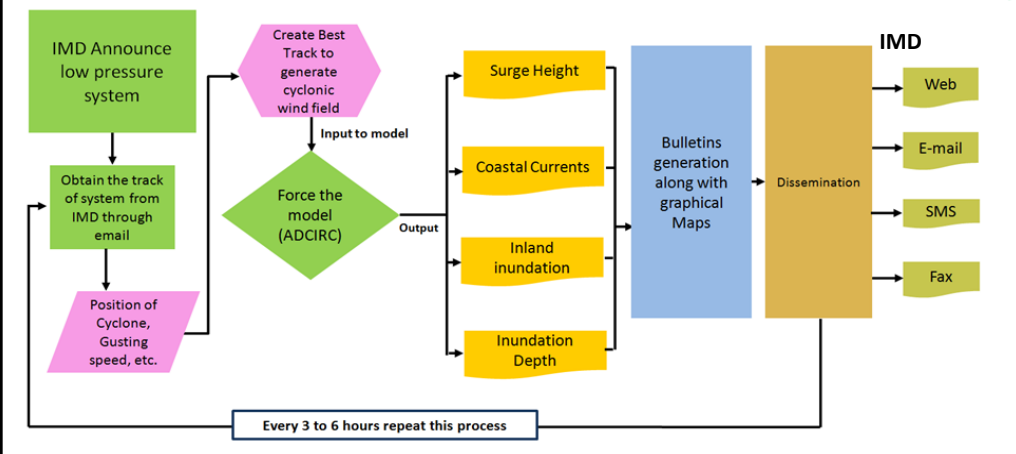
Storm Surge Early Warning Architecture



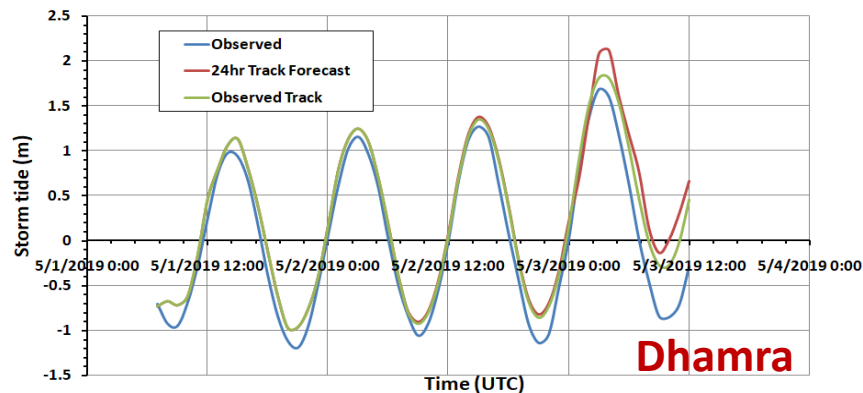
Storm Surge Early Warning System at INCOIS

- INCOIS has set-up the storm surge early (SSEWS) warning system for the Indian coasts using ADCIRC model.
- SSEWS utilizes the automated Decision Support System (DSS) based on Geographic Information System (GIS) and database technology.

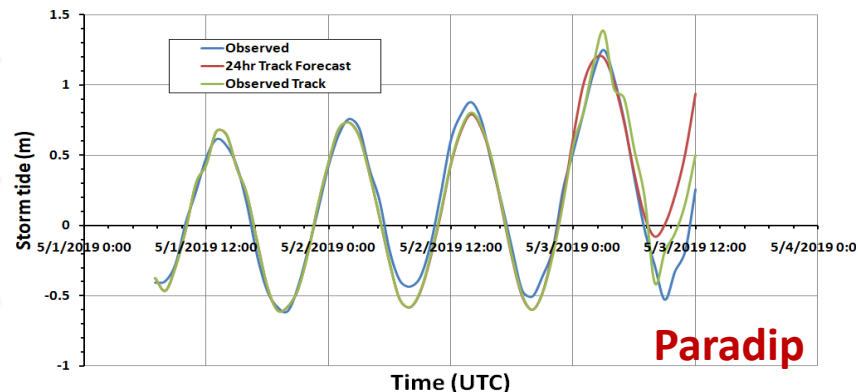
Standard Operating Procedure



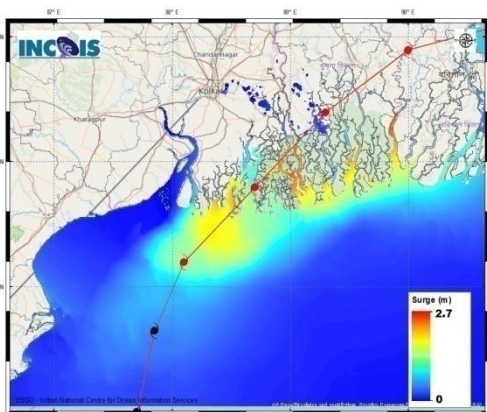
Storm surge validation against the tide gauge records during Hudhud



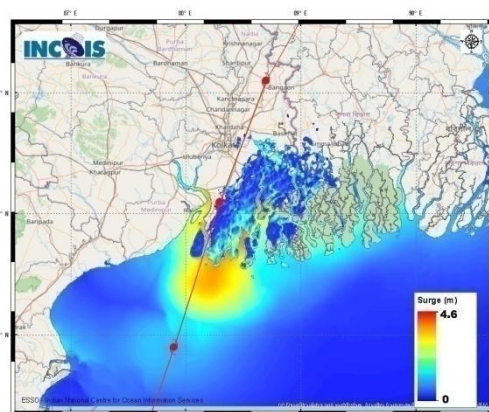
Storm surge validation against the tide gauge records during Fani



Storm surge validation against the tide gauge records during Fani



Storm surge map during 'Bulbul'



Storm surge map during 'Amphan'

Best Track Conversion to Parametric wind model

Sub: Cyclonic storm, PHAILIN in East central Bay of Bengal:
Cyclone Alert for North Andhra Pradesh and Orissa Coast. Cyclone Warning for
Andaman & Nicobar Island

The deep depression over east central Bay of Bengal remained practically stationery, intensified into a cyclonic storm, **PHAILIN** and lay centred at 1730 hrs IST of today, 09th October 2013 over near latitude 13.5°N and longitude 92.5°E, about 220 km north-northwest of Port Blair, 950km southeast of Paradip, 1100 km east-southeast of Visakhapatnam. The system would intensify into a severe cyclonic storm during next 24 hours. It would continue to move west-northwestwards for some time and then north-westwards and cross north Andhra Pradesh and Odisha coast between Kalingapatnam and Paradip by night of 12th October, 2013 as a very severe cyclonic storm with a maximum sustained wind speed of 175-185 kmph.

Based on latest analysis with NWP models and other conventional techniques, estimated track and intensity of the system are given in the Table below:

Date/Time(IST)	Position (Lat. °N/ Long. °E)	Sustained maximum surface wind speed (kmph)	Category
09-10-2013/1730	13.5/92.5	65-75 gusting to 85	Cyclonic Storm
09-10-2013/2330	13.8/92.0	70-80 gusting to 90	Cyclonic Storm
10-10-2013/0530	14.3/91.0	75-85 gusting to 95	Cyclonic Storm
10-10-2013/1130	14.7/90.4	85-95 gusting to 105	Cyclonic Storm
10-10-2013/1730	15.2/89.8	95-105 gusting to 120	Severe Cyclonic Storm
11-10-2013/0530	15.7/88.3	115-125 gusting to 140	Severe Cyclonic Storm
11-10-2013/1730	16.5/87.1	135-145 gusting to 160	Very Severe Cyclonic Storm
12-10-2013/0530	17.3/86.0	155-165 gusting to 180	Very Severe Cyclonic Storm
12-10-2013/1730	18.5/85.2	175-185 gusting to 200	Very Severe Cyclonic Storm
13-10-2013/0530	20.0/84.7	70-80 gusting to 90	Cyclonic Storm
13-10-2013/1730	21.2/84.4	50-60 gusting to 70	Deep Depression
14-10-2013/0530	22.0/84.4	35-45 gusting to 55	Depression

IMD input Data

INPUT

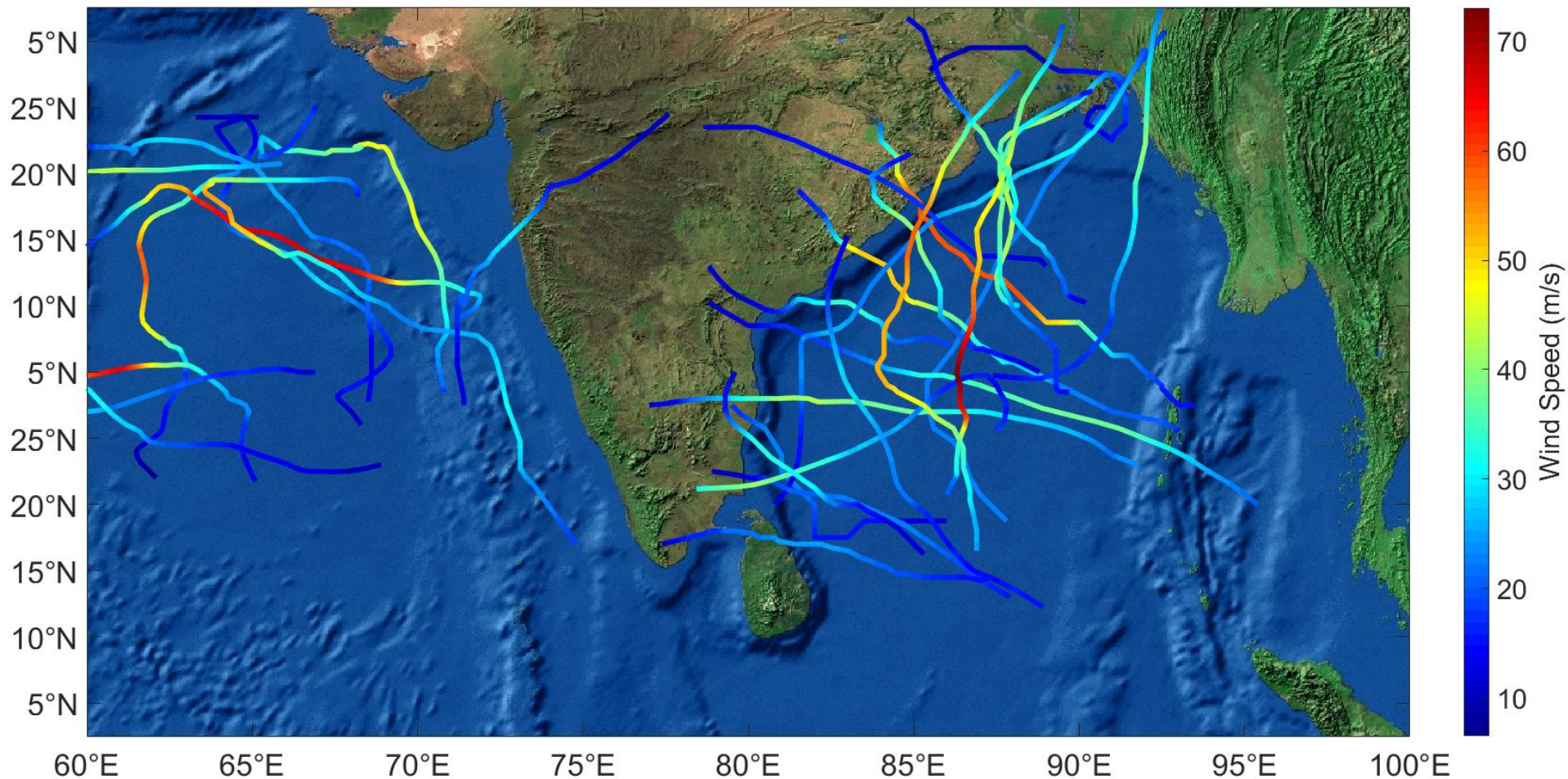
Under the influence of this system, rainfall at most places with heavy to very heavy falls at a few places and isolated extremely heavy falls (≥ 25cm) would occur over Andaman and Nicobar Islands during next 12 hrs. Isolated heavy to very heavy rainfall would occur in subsequent 24 hrs.

Best Track Data input To parametric wind model

OUTPUT

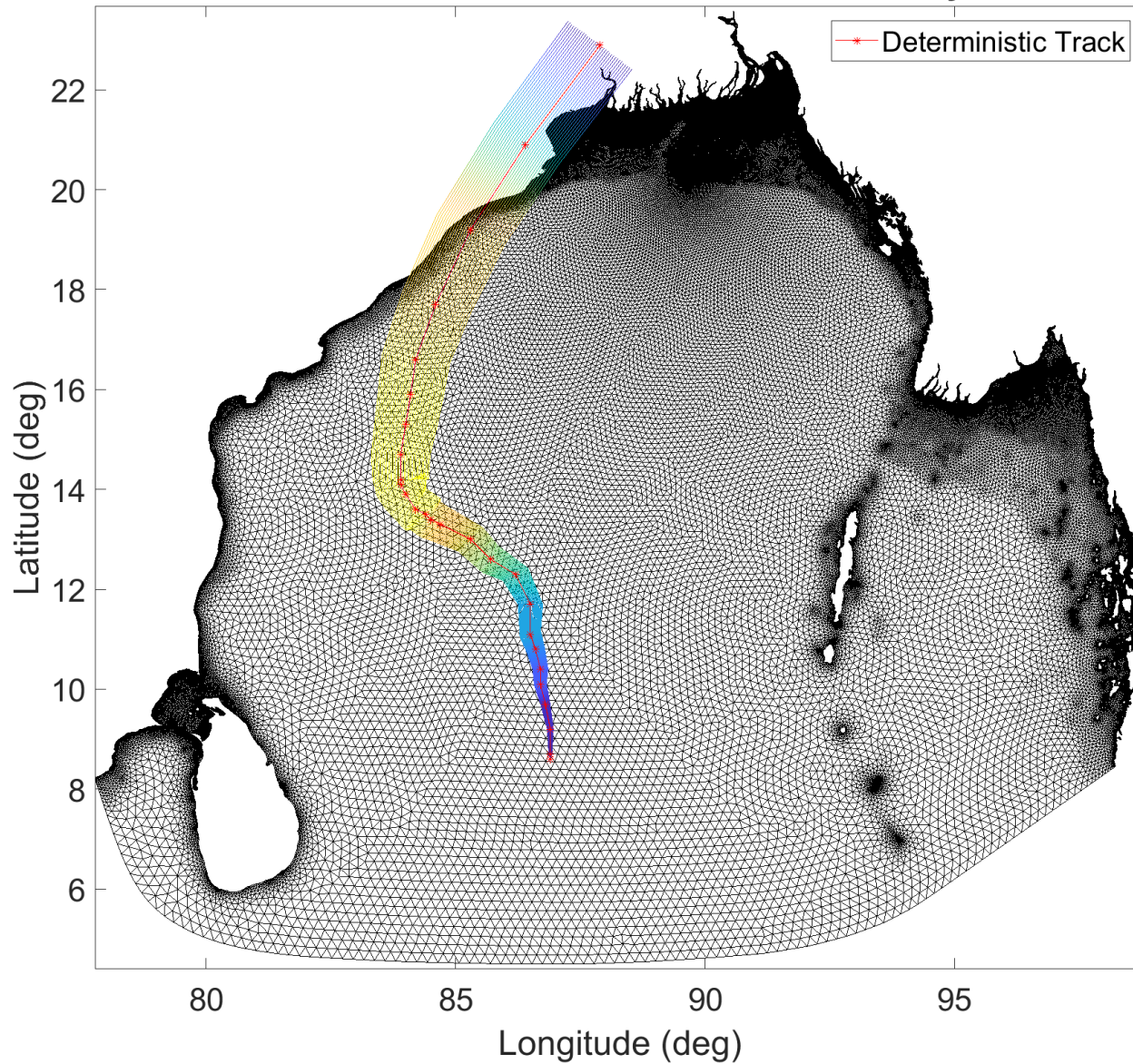
CYCLONE TRACK				
TRACK ISSUED BY IMD				
12				
13.5	92.5	35000	8.0	0
13.8	92.0	35000	9.0	6
14.3	91.0	35000	10.0	6
14.7	90.4	35000	13.0	6
15.2	89.8	30000	16.0	6
15.7	88.3	30000	22.0	12
16.5	87.1	30000	30.0	12
17.3	86.0	25000	39.0	12
18.5	85.2	25000	49.0	12
20.0	84.7	35000	9.0	12
21.2	84.4	40000	5.0	12
22.0	84.4	45000	4.0	12

Tracks of the cyclonic systems (38) that were monitored and issued real-time storm surge warnings since 2013



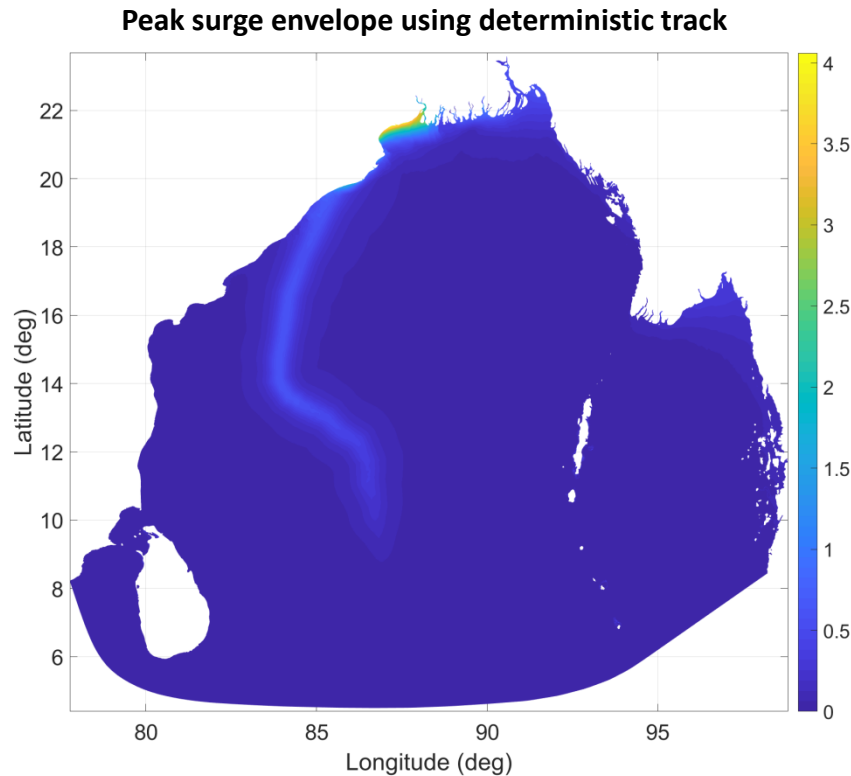
Probabilistic Storm Surge Estimation

Shifted tracks with in cone of uncertainty

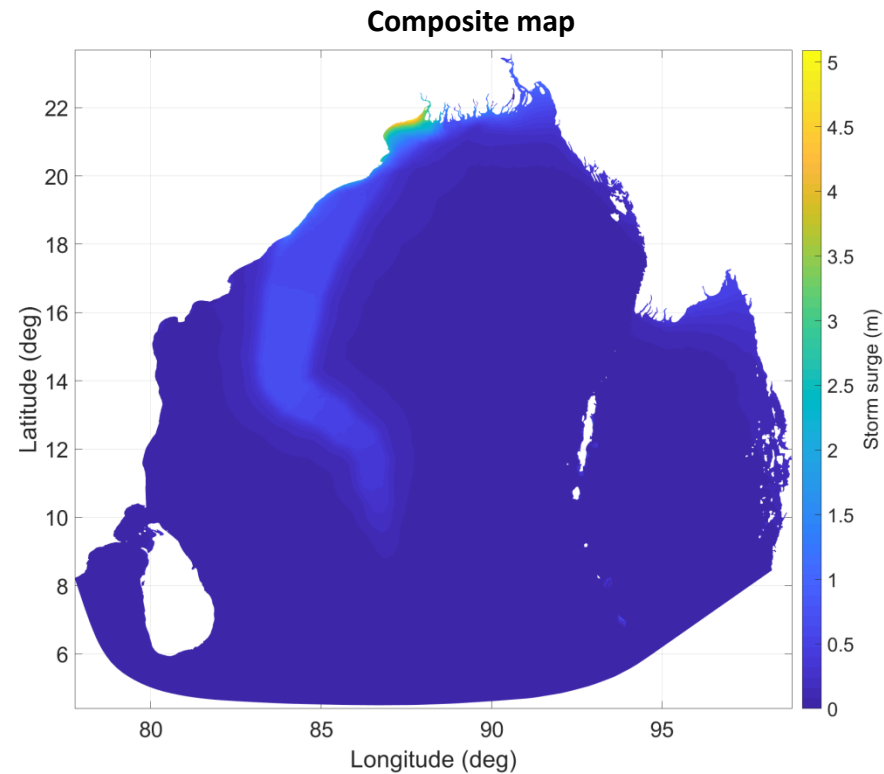


Ongoing enhancements

Representation of peak surge envelope from deterministic track and the respective composite map from all the tracks

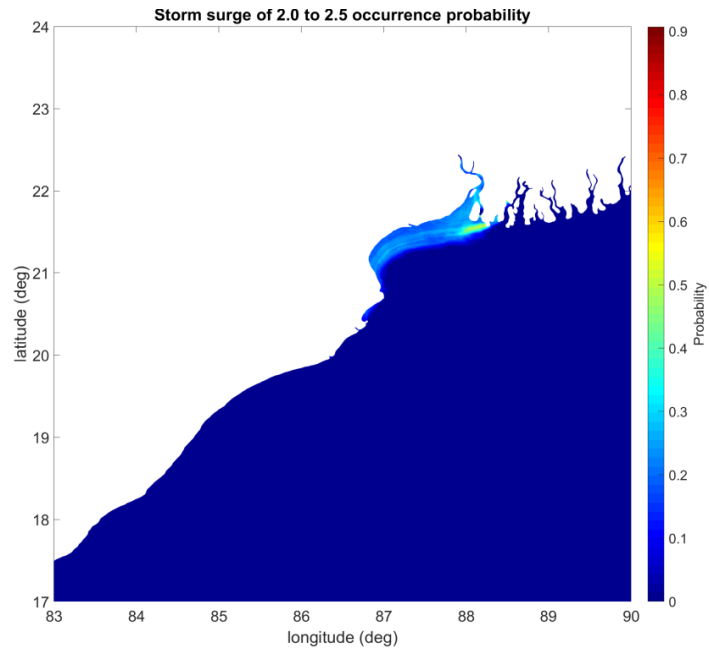
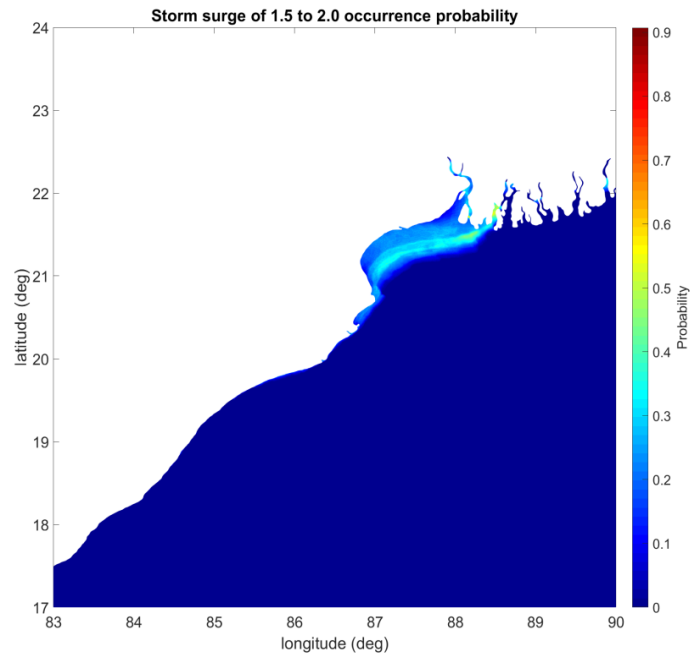
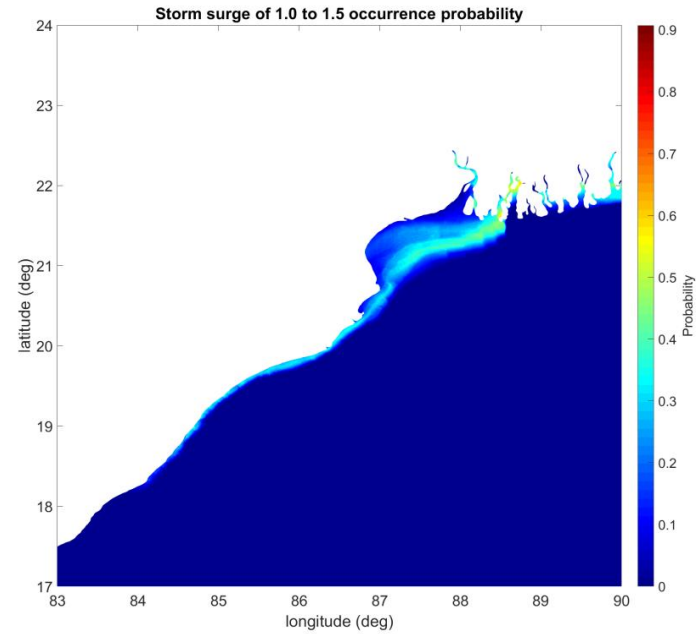
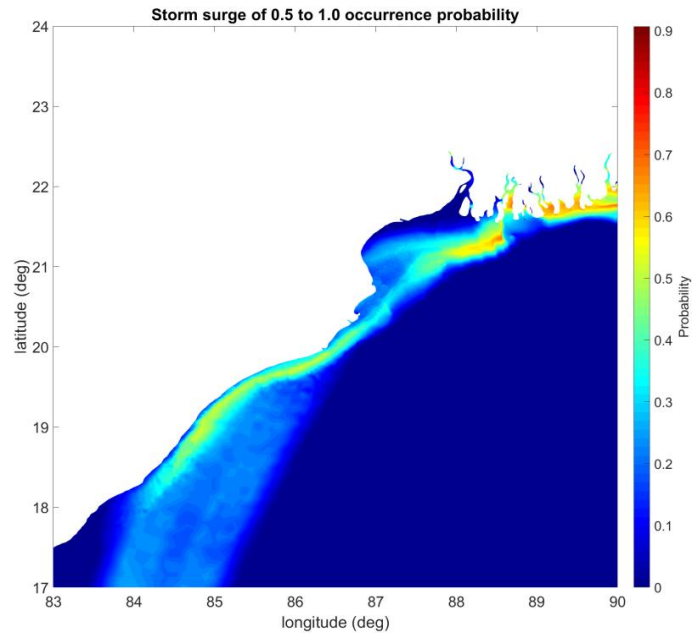


Single track



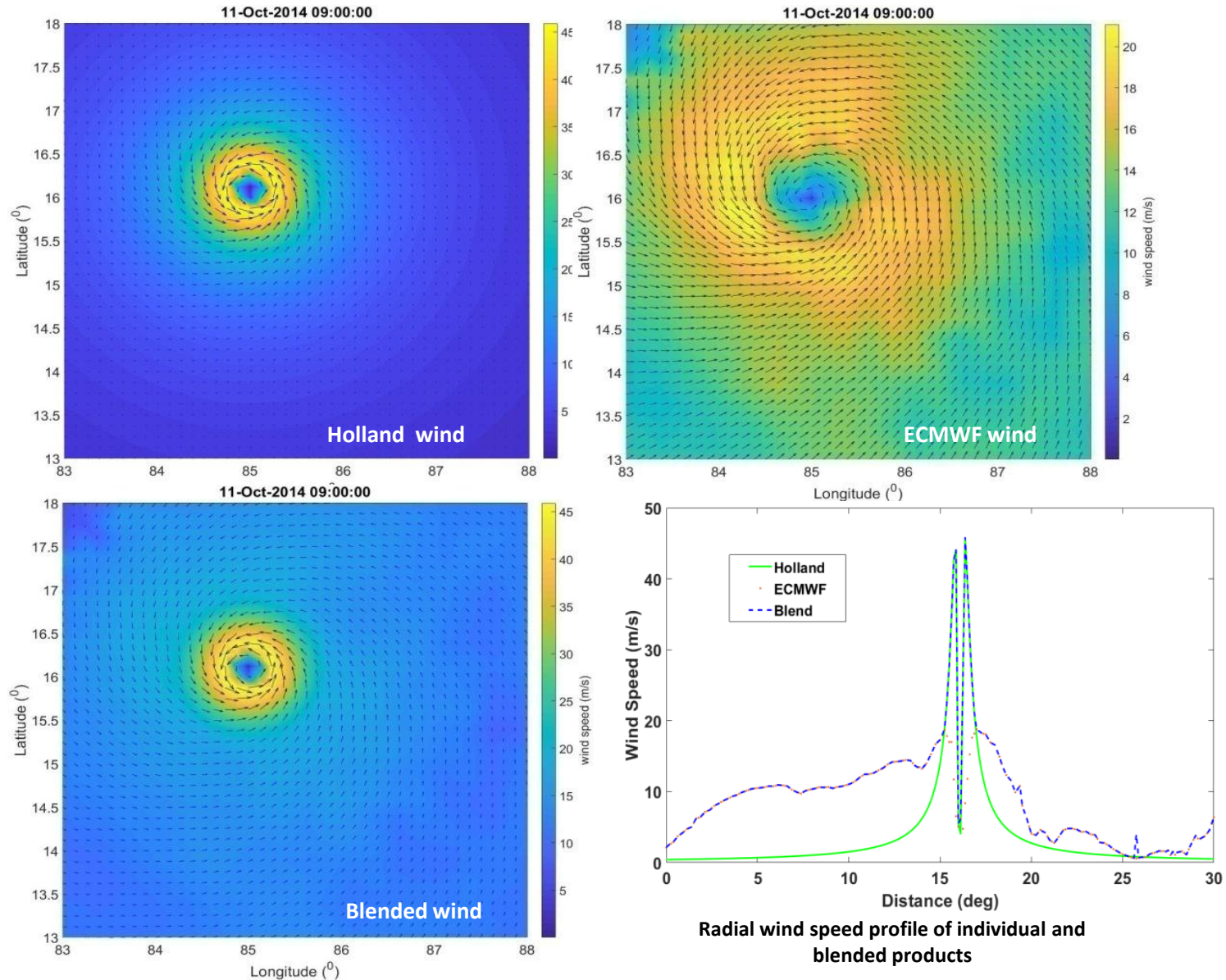
Multiple tracks (~ 100)

Storm surge occurrence probability for a given range



Blending of parametric and global wind speeds

Snap shot of three different wind fields



Ongoing enhancements

Parametric wind Vs. Blended wind

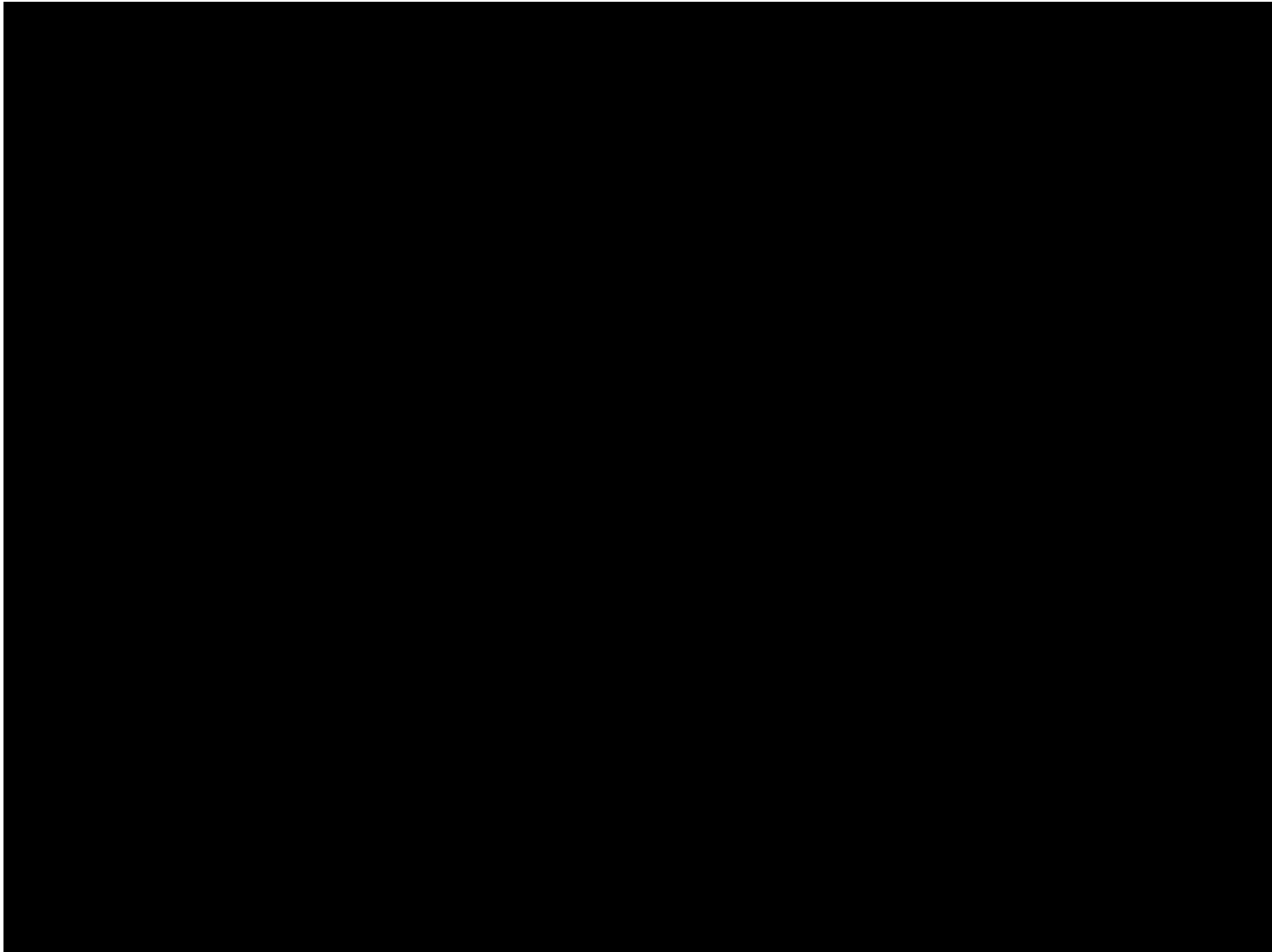
Ongoing enhancements

Titli cyclone (October, 2018)						
Buoy	CC		RMSE (m/s)		SI (%)	
	Parametric	Blended	Parametric	Blended	Parametric	Blended
BD 08	0.17	0.89	6.6	2.0	53	16
BD 09	-0.06	0.76	6.3	2.1	53	18
BD 10	0.13	0.65	3.5	2.5	35	28
Hudhud cyclone (October, 2014)						
Buoy	CC		RMSE (m/s)		SI (%)	
	Parametric	Blended	Parametric	Blended	Parametric	Blended
BD 10	0.58	0.80	5.7	1.3	44	9
BD 11	0.52	0.85	4.5	2.5	45	25
BD 13	0.37	0.42	9.0	3.3	74	27
Phailin cyclone (October, 2014)						
Buoy	CC		RMSE (m/s)		SI (%)	
	Parametric	Blended	Parametric	Blended	Parametric	Blended
BD 08	0.86	0.92	5.8	1.6	40	12
BD 09	0.83	0.83	5.2	2.5	38	18
BD 11	0.81	0.90	3.0	1.4	39	18
Thane cyclone (December, 2011)						
Buoy	CC		RMSE (m/s)		SI (%)	
	Parametric	Blended	Parametric	Blended	Parametric	Blended
BD11	0.67	0.95	2.78	1.57	21	12
BD13	0.96	0.98	2.47	1.74	20	14

Woodcock et al. 2007: SI should be less than 30%.

Tsunami

How Tsunami occurs?



Source: <https://youtu.be/APDvwIL8jZk>

TSUNAMI

Propagation of disturbance on the ocean free surface under gravitational force from the source to coast.

TSUNAMIS GENERATED BY

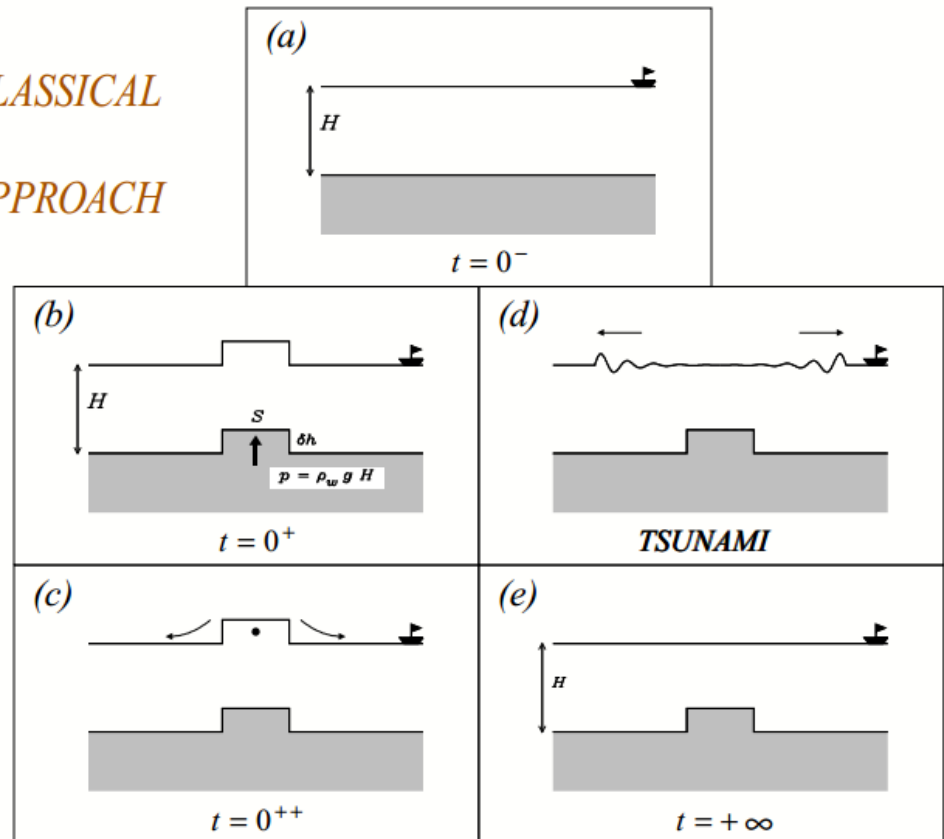
- ❖ Earthquakes
- ❖ Landslides
- ❖ Volcanic Explosions
- ☐ Meteo – Tsunamis

Tsunami Simulation Steps

1. Generation
2. Propagation
3. Run Up/ Inundation

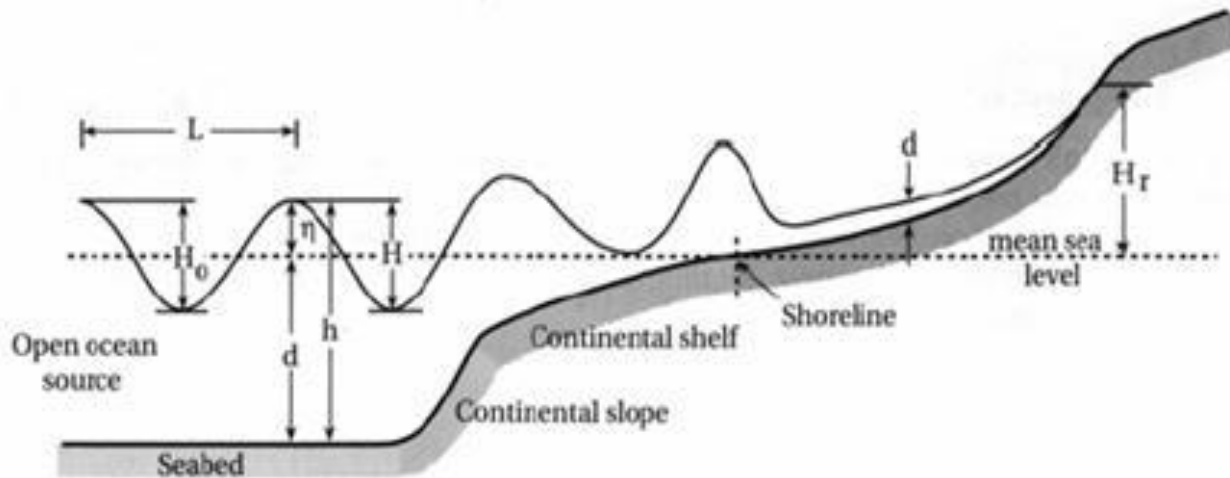
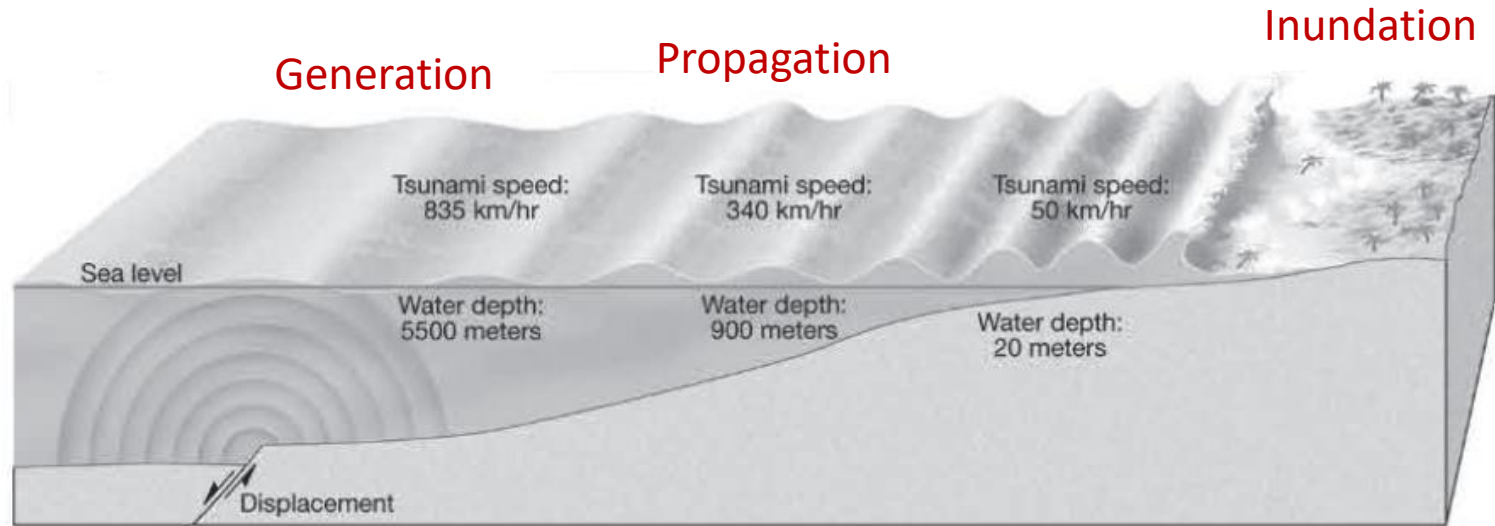
CLASSICAL

APPROACH



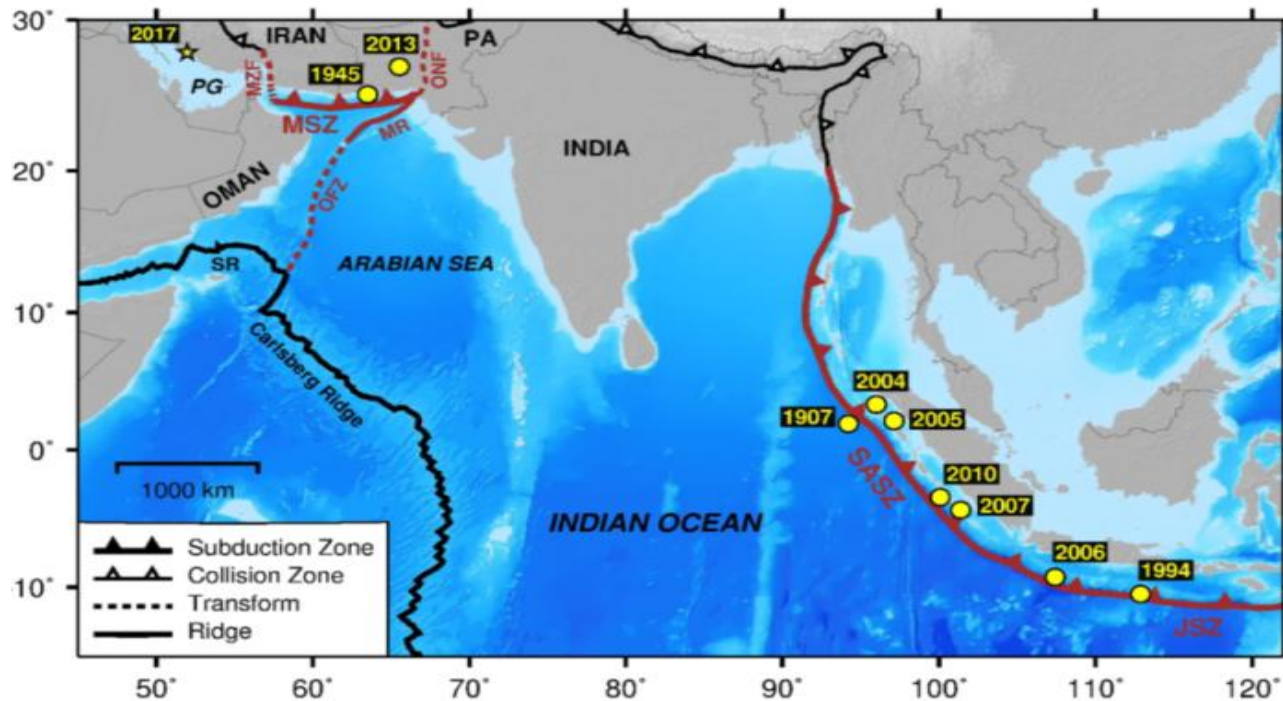
Classical Approach: Source – Prof. Emile A. Okal, Northwestern University, 2017.

Tsunami characteristics and definitions



L	:	wavelength (m)
H_0	:	wave height at deep water (m)
d	:	water depth (m)
η	:	wave elevation (m)
h	:	water elevation (m)
H	:	wave height (m)
H_r	:	wave run-up (m)

Potential Tsunamigenic Zones in the Indian Ocean



- Tsunamis are primarily caused due to large undersea Earthquakes.
- For a tsunami to hit Indian coast, it is necessary that a tsunamigenic earthquake occurs and its magnitude should be larger than M 7. Possible locations of such events are enclosed in ellipse
- Earthquakes with Slow Rupture Velocities are most efficient Tsunami Generators
- 75% of earthquake energy is released in the circum-Pacific belt – 900 Tsunamis in 20th Century
- 20% in the Alpine-Himalayan belt – 6 Tsunamis in 20th Century

- **Historical Tsunami in Indian Ocean**

12 Apr, 1762 (BoB EQ) – 1.8 M

31 Dec, 1881 (Car Nicobar EQ)

27 Aug, 1883 (Krakatoa) – 2 M

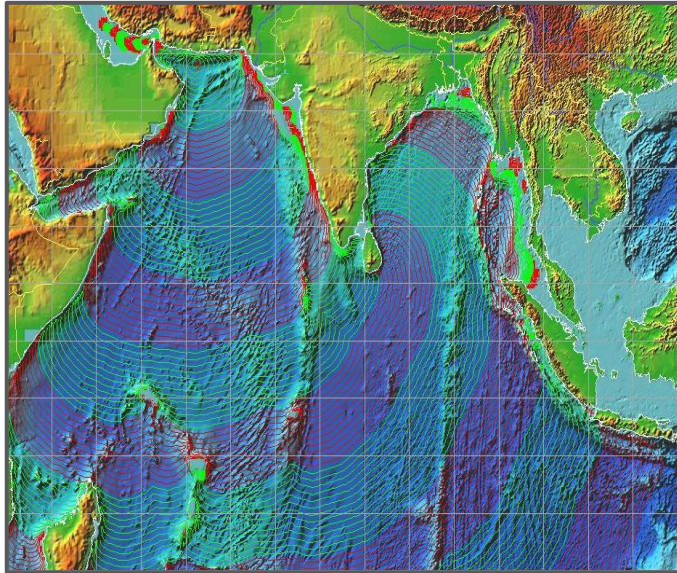
26 Jun, 1941 (Andaman EQ)

27 Nov, 1945 (Makran EQ) – 12 M

26 Dec, 2004 (Sumatra EQ)

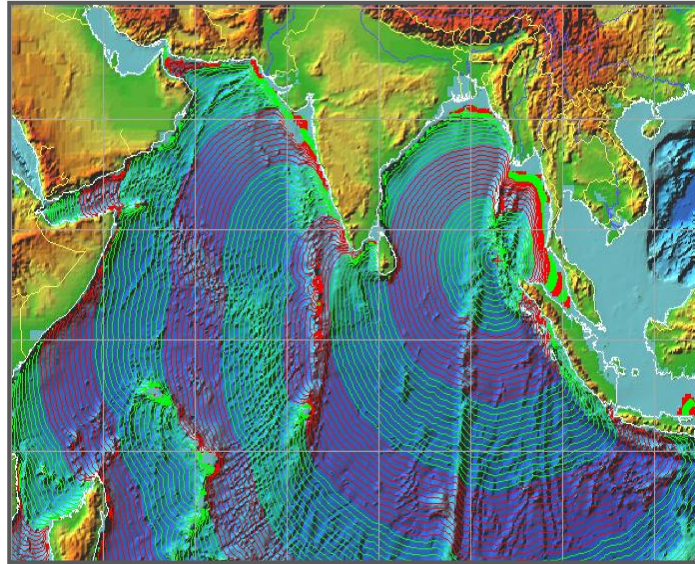
Tsunami Risk Assessment

Makran Subduction Zone



- If Earthquake occurs at Makran Subduction zone, Travel Time to nearest Indian Coast (Gujarat) are 2 to 3 hrs

Andaman-Sumatra Subduction Zone



- If Earthquake happens at Nicobar Islands , travel times to nearest coast (A&N Islands) are 20 to 30 min
- For Indian main land travel times are 2 to 3 hrs

Tsunami Travel Times & Response time

- Depending upon the Earthquake location (Makran/Andaman-Sumatra Subduction Zone) the response time for evacuation of coastal population could range between 10 min to few hours.
- As Andaman & Nicobar Islands situated right on subduction zone the available response time is very short

SOP – Public Response and Threat Levels in Bulletins

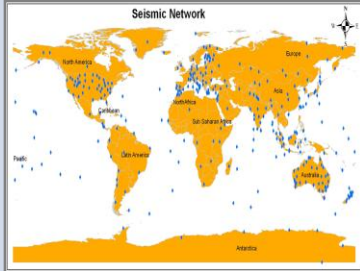
	Threat Status	Action to be taken	Dissemination to		
> 2.0m	WARNING	Public should be advised to move inland towards higher grounds. Vessels should move into deep Ocean	MoES, MHA, NDMA, NCMC, NDRF Battalions, SEOC, DEOC, Public, Media	WARNING	
0.5m-2.0m	ALERT	Public should be advised to avoid beaches and low-lying coastal areas. Vessels should move into deep Ocean	MoES, MHA, NDMA, NCMC, NDRF Battalions, SEOC, DEOC, Public, Media	ALERT	
> 0.5m	WATCH	No immediate action is required	MoES, MHA, NDMA, NCMC, NDRF Battalions, SEOC, DEOC, Media	WATCH	
< 0.5m	THREAT PASSED	All clear determination to be made by the local authorities	MoES, MHA, NDMA, NCMC, NDRF Battalions, SEOC, DEOC, Public, Media	THREAT PASSED	

Components of Tsunami Early Warning System

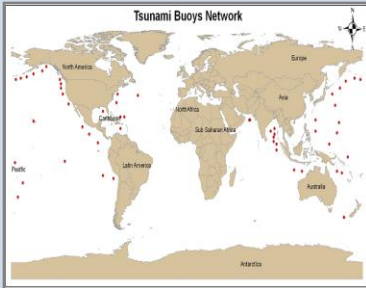
Detection

Warnings

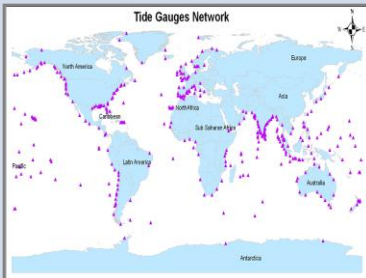
Dissemination



Seismic Network



BPR Network



Tide gauge Network

Observation Networks



VSAT



INSAT

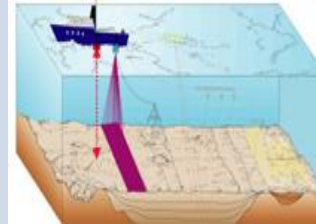


GPRS

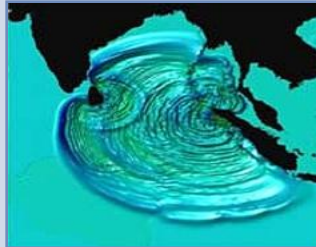


INMARSAT

Communications



Bathymetry



Tsunami Modelling



Topography



Costal Vulnerability

Simulations

Participating Institutions
IMD, NIOT, ICMAM, SOI,
ISRO, NRSC, INCOIS
MHA, NDMA, Coastal States



Capacity Building



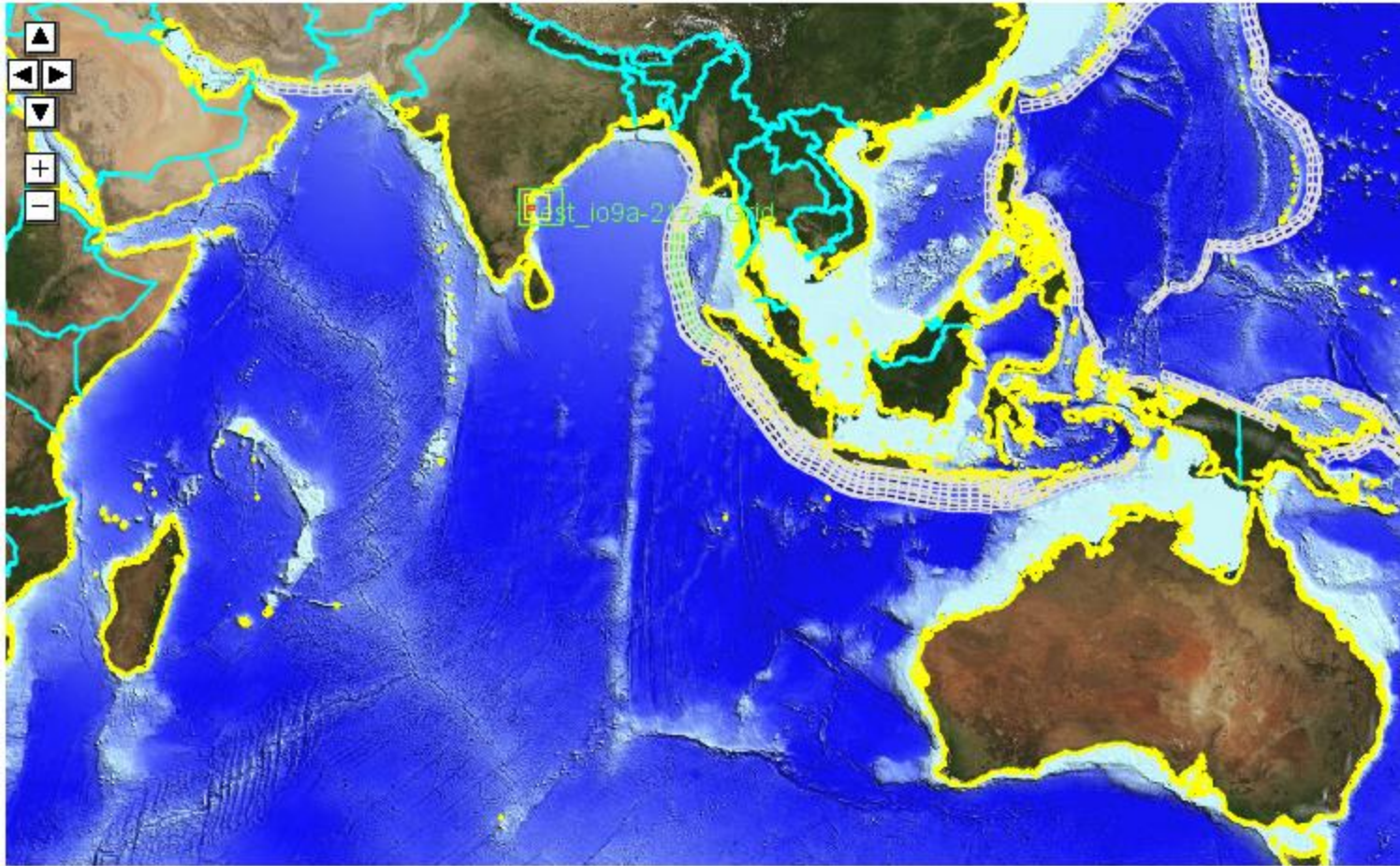
R & D

Last mile connectivity

**TSUNAMI
WARNINGS!!!**

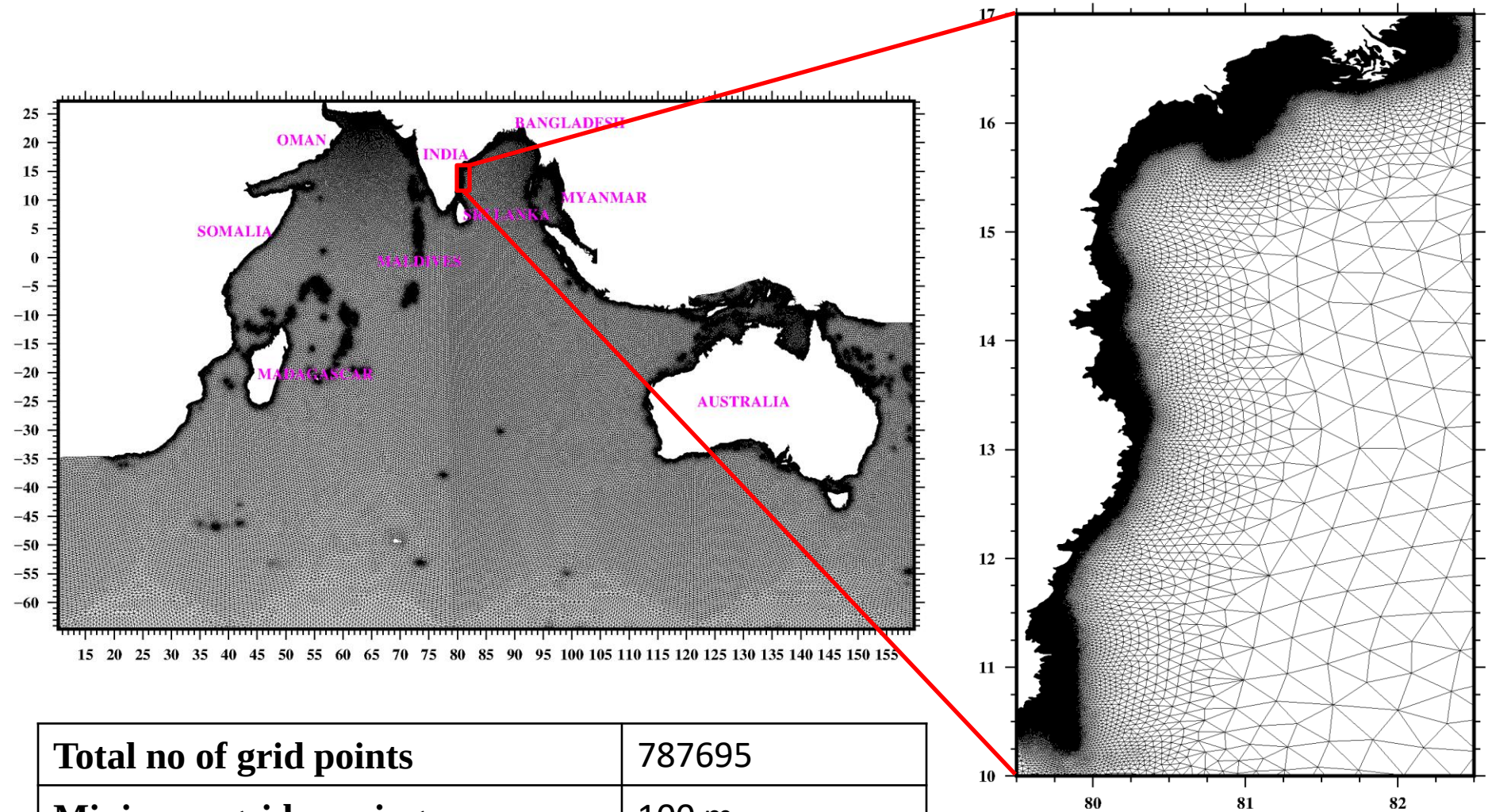


Tsunami modeling using Pre defined unit sources and pre computed scenarios



Model: Tunami

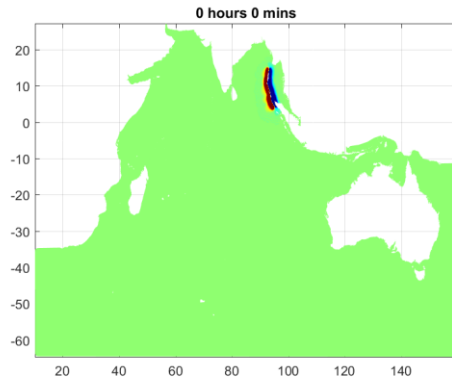
Modeling tsunami and its associated inland inundation using ADCIRC



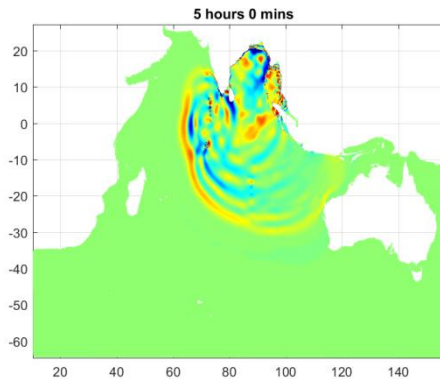
Total no of grid points	787695
Minimum grid spacing	100 m
Maximum grid spacing	30 km

Stages of Tsunami Modeling

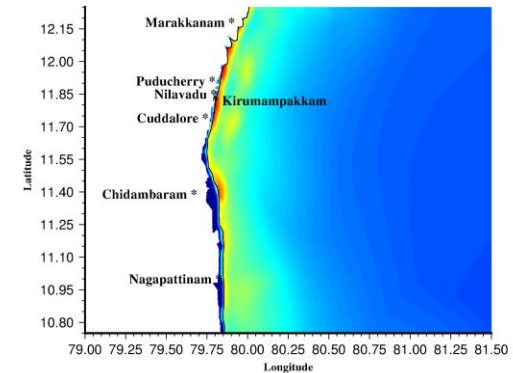
Generation



Propagation



Run up/ Inundation



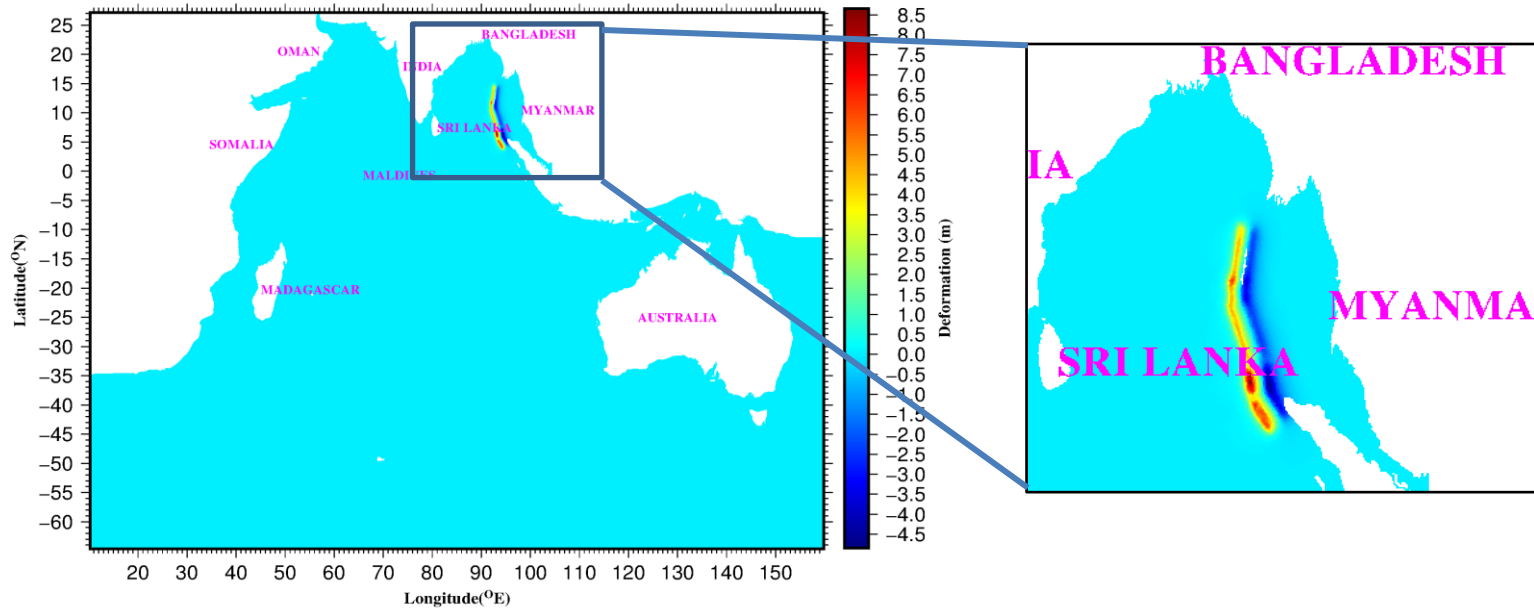
Inputs required

1. Bathymetry (propagation) and topography (inundation) grid for a region covering subduction zone
2. Seismic fault parameters to generate initial deformation
(longitude, latitude, depth, faults length, faults width, strike angle, dip angle, rake angle and slip)

Outputs obtained

1. Wave heights at every grid/ specified points for a specified time step
2. Velocity components at every grid/ specified points for a specified time step

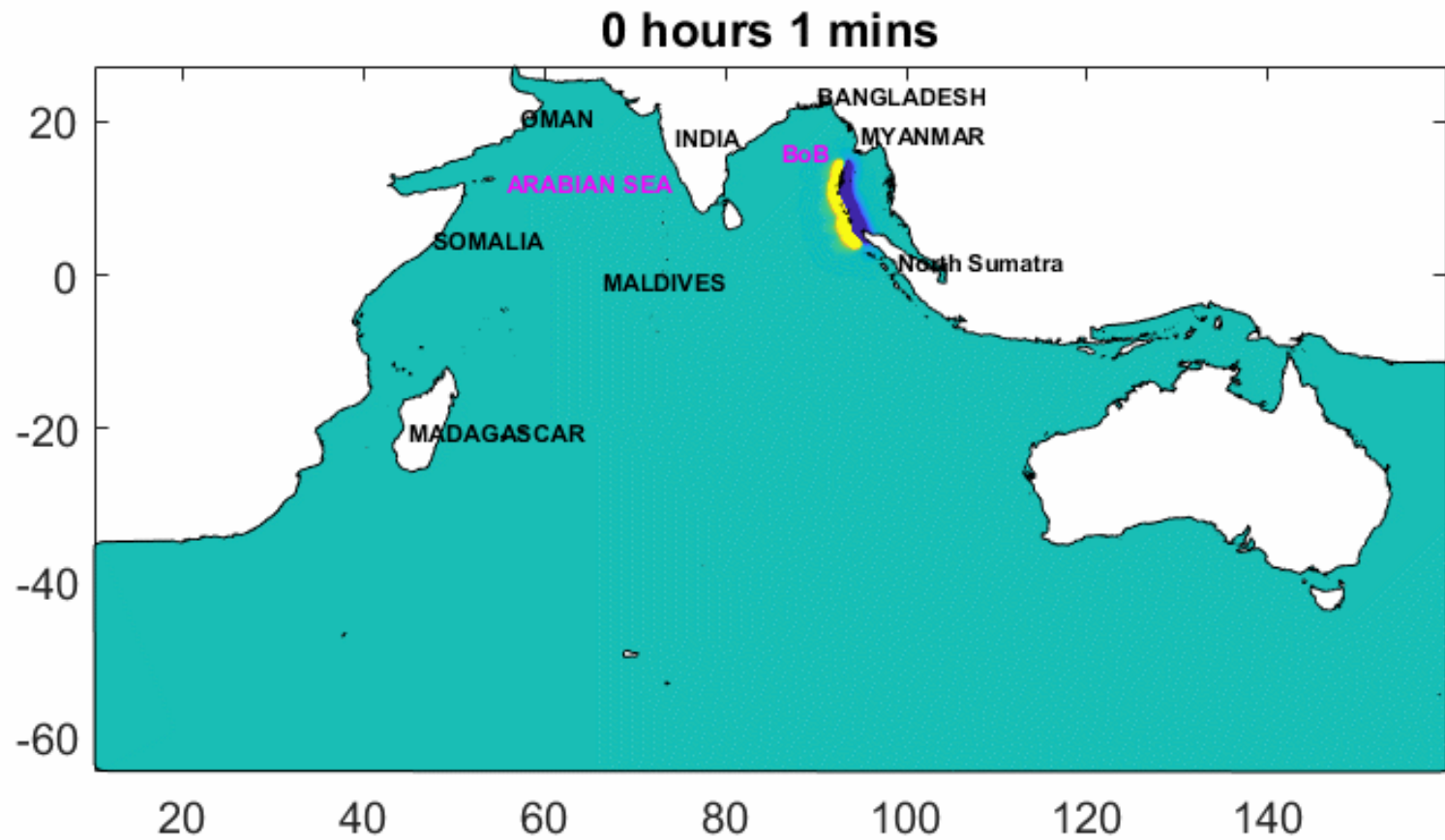
Generation



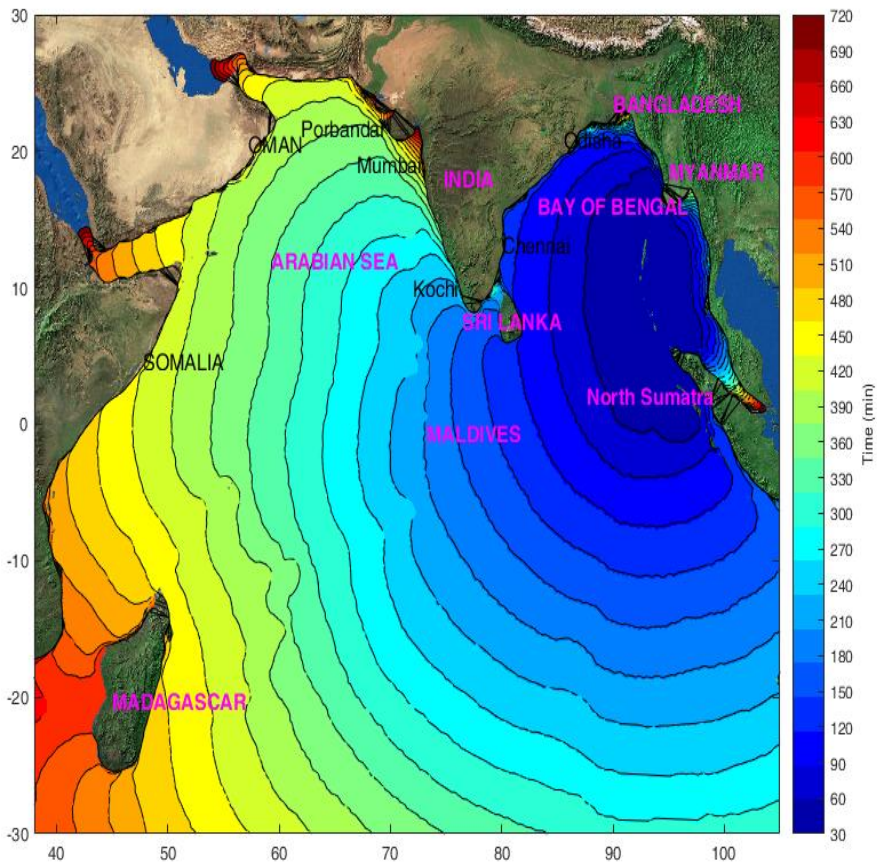
Parameters	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5
x_o (longitude)	94.57	93.90	93.21	92.60	92.87
y_o (latitude)	3.83	5.22	7.41	9.70	11.70
d (km)	25	25	25	25	25
ϕ (degrees)	323	348	338	356	10
λ (degrees)	90	90	90	90	90
δ (degrees)	12	12	12	12	12
Δ (m)	18	23	12	12	12
L (km)	220	150	390	150	350
W (km)	130	130	120	95	95

1. Mansinha, L. and Smylie, D.E, 1971. "The displacement fields of inclined faults" Bulletin of the Seismological Society of America, Vol. 61, 1433-1440.
2. Grilli, S. T., Ioualalen, M., Asavanant, J., Shi, F., Kirby, J., and Watts, P.: Source constraints and model simulation of the December 26, 2004 Indian Ocean tsunami, J. Waterway Port Coastal and Ocean Engineering, 133(6), 414–428, 2007.

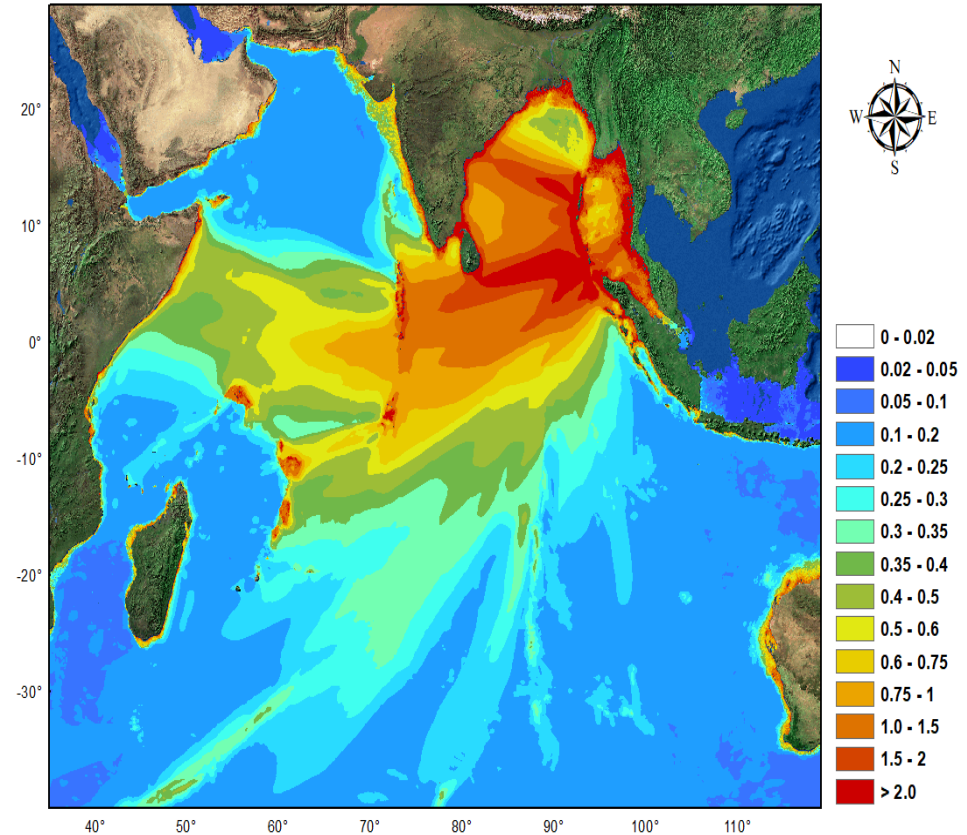
Propagation



Propagation....

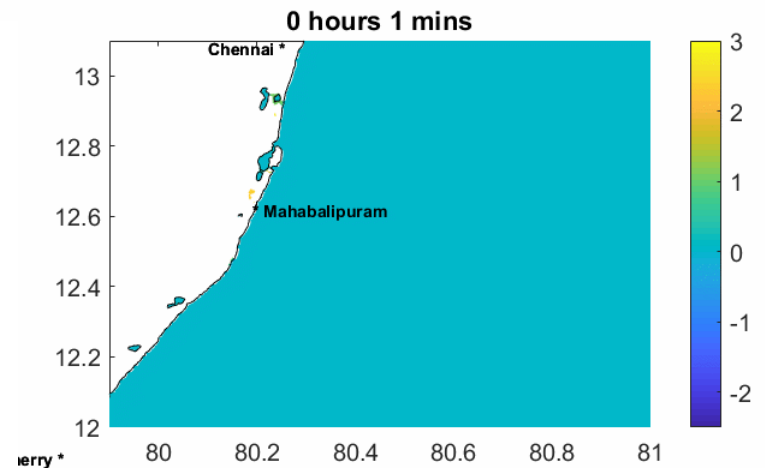
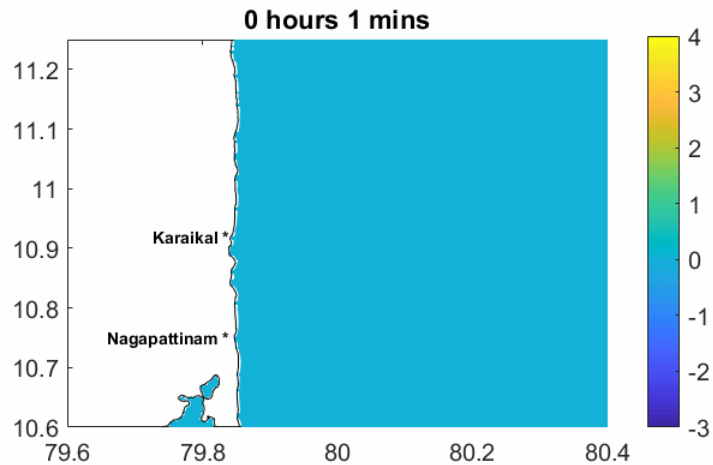
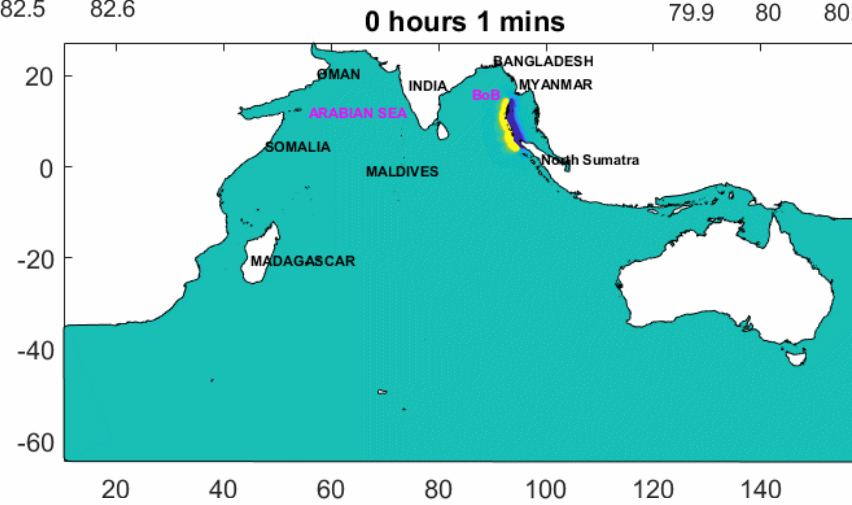
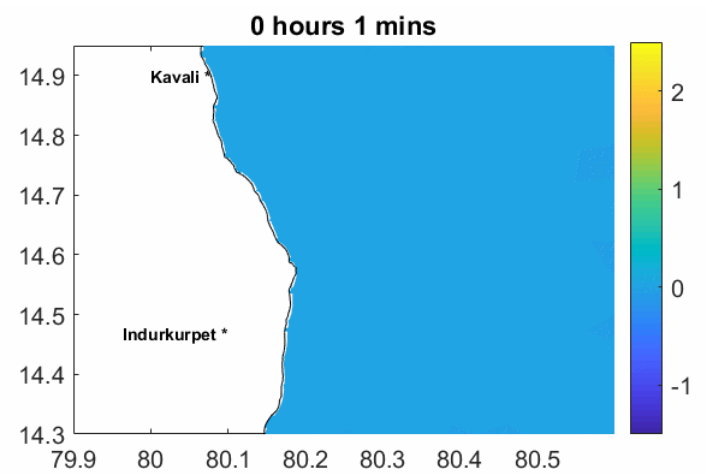
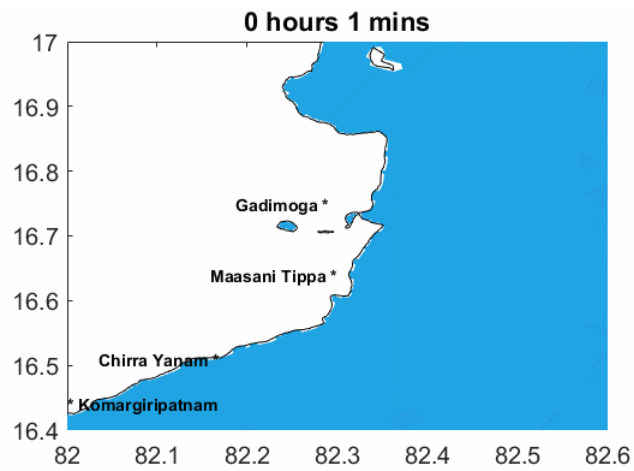


**Travel Times
(One Hour
Interval)**

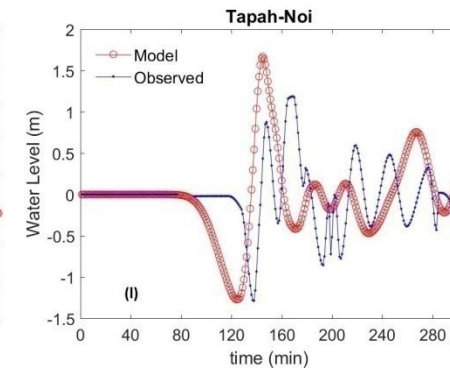
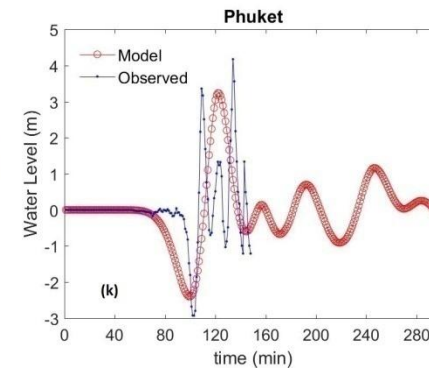
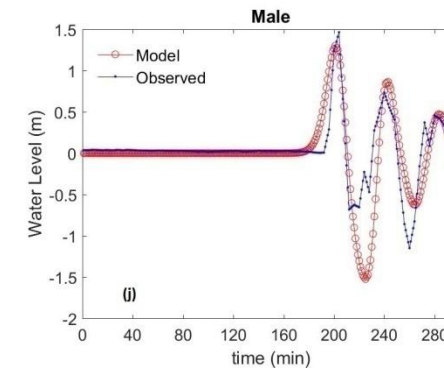
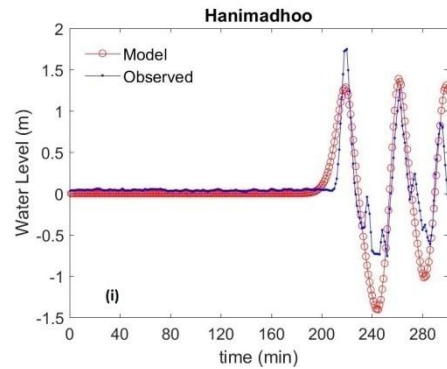
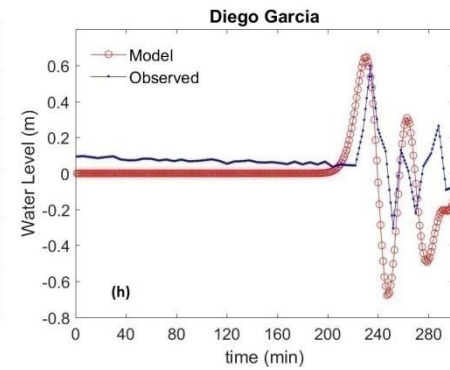
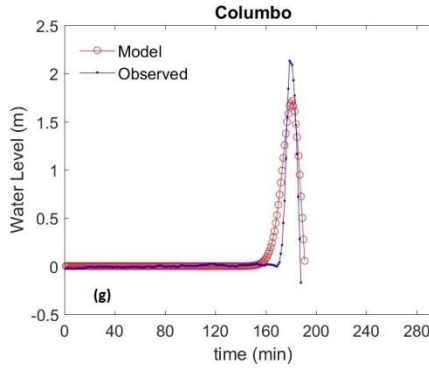
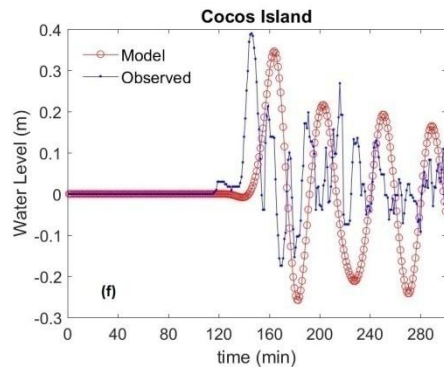
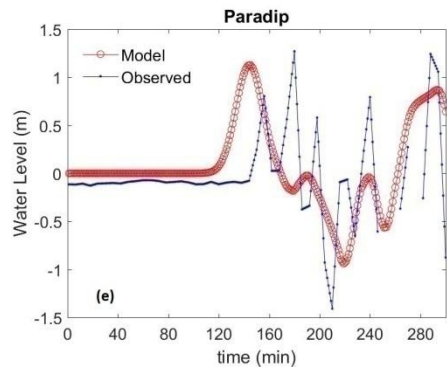
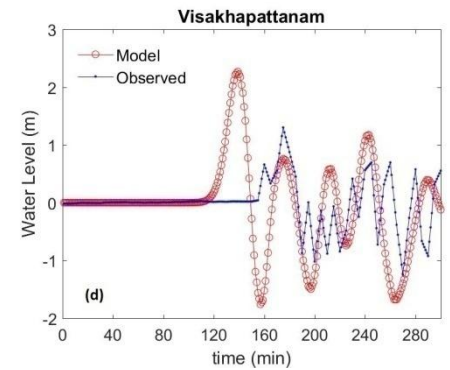
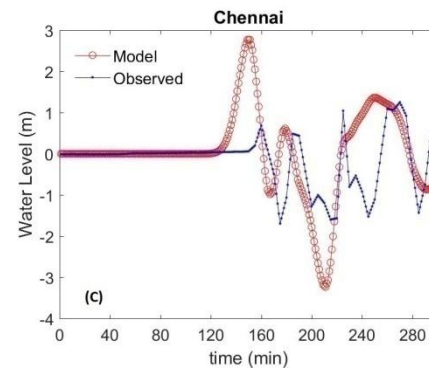
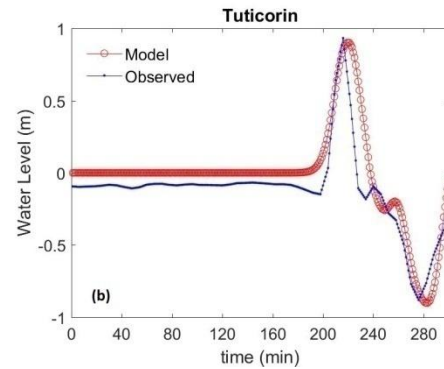
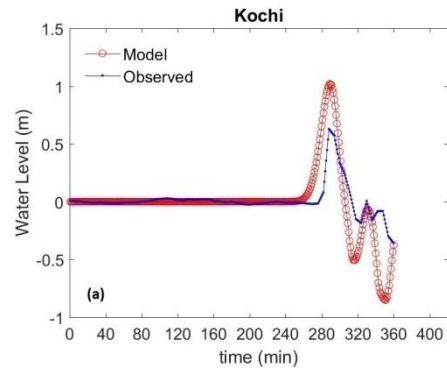


**Directivity Plot
(Wave height in
meters)**

Inundation



Comparison with observations



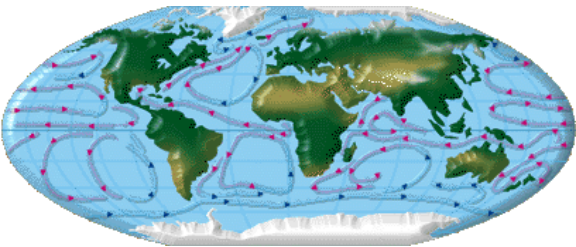
Wave

Ocean Forecast- the parameters



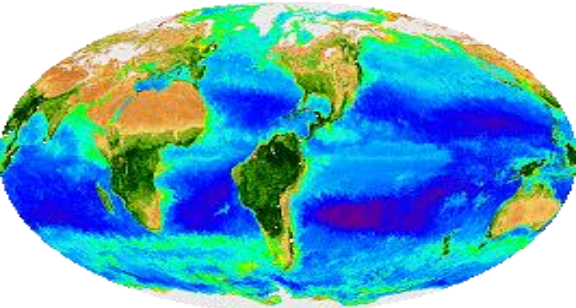
Sea State/ Marine-
met/Tectonic

High Wave, Swell, Wind,
Storm surge, Tsunami



General Circulation

Surface current, SST, T&S profile,
mixed layer depth, thermocline,
undercurrents, tides



Biogeochemical/
Ecological

Nutrients, Chlorophyll,
Oxygen, productivity,
Fishery, HAB



Marine Pollution

Oilspill, Pollutants etc.

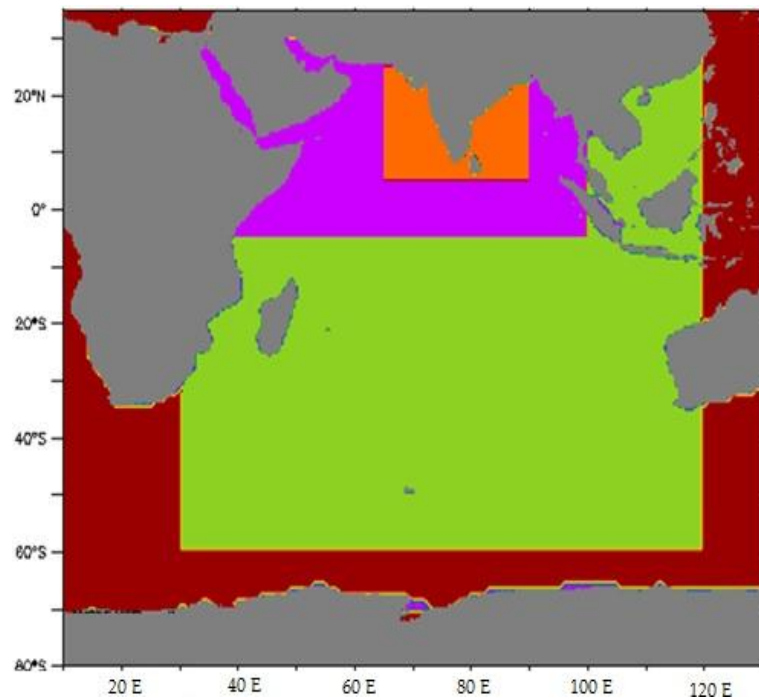
WAVEWATCHIII Multi-Grid setup at INCOIS

Objective: Improvement of wave prediction at Global as well as regional scales with special emphasis on Indian Ocean.

➤ WAVEWATCHIII-V4.18 is operational and provides 10 days wave forecast in advance using two different wind forcing, NCMRWF and ECMWF.

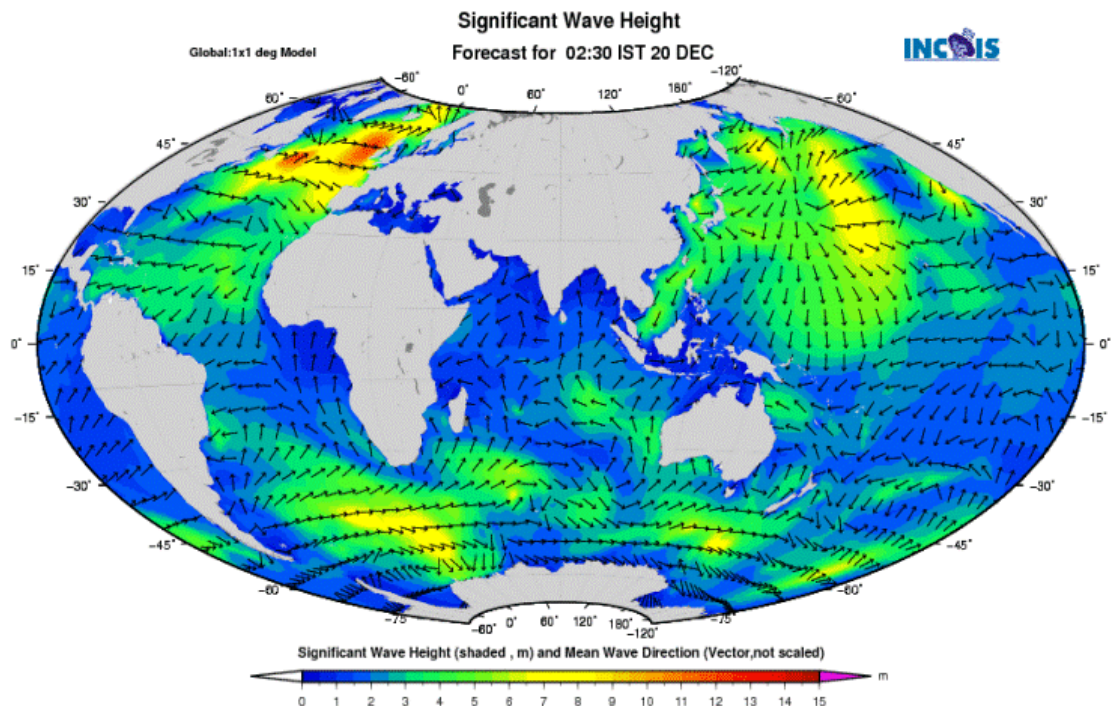
➤ OSF-Regional wave forecast (Arabian Sea, Bay of Bengal, Persian Gulf, Red Sea, South China Sea, Southern Indian Ocean, Northern Indian Ocean) is provided using WWIII forecasted wave parameters.

➤ This set-up also provides boundary conditions for the coastal wave models such as SWAN and MIKE21 SW for the coastal wave forecast.



- Spatial resolution
 - 1 deg Southern Hemisphere grid
 - 0.5 deg Indian Ocean grid
 - 0.25 deg North Indian Ocean grid
 - 0.04 deg Coastal grid

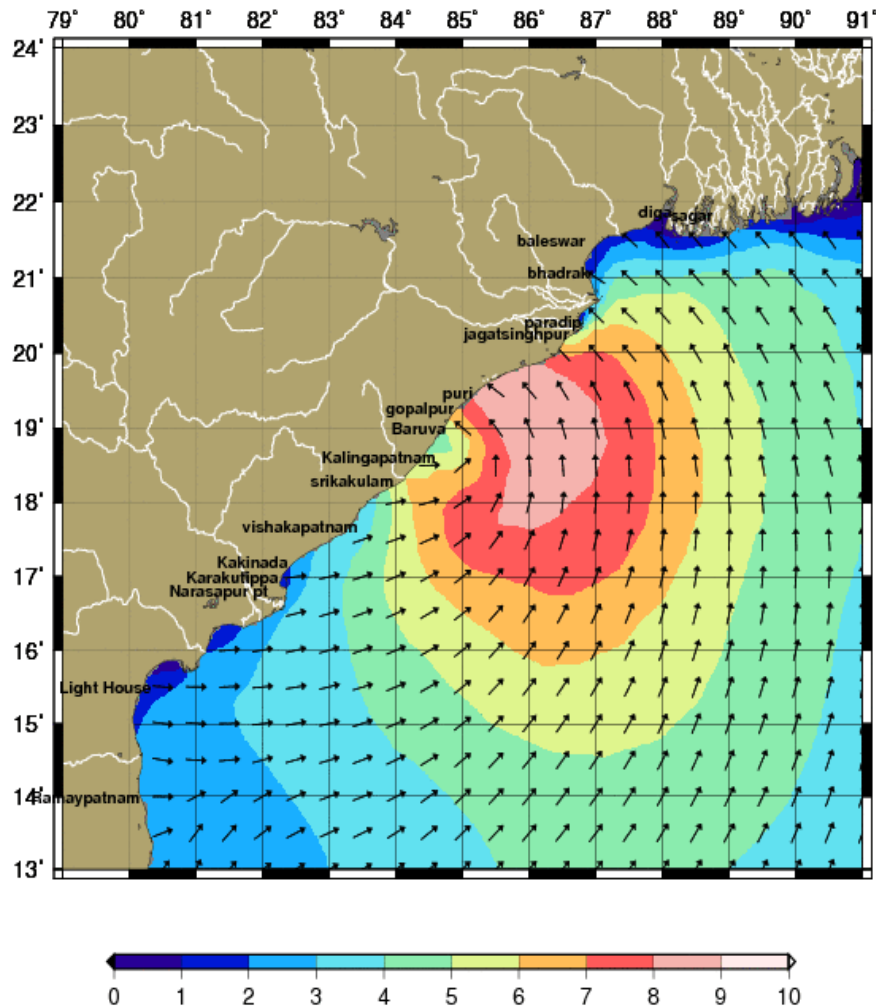
- Bathymetry- Etopo1



Wave forecast during a cyclone

Significant Wave height (m) and Direction (°)

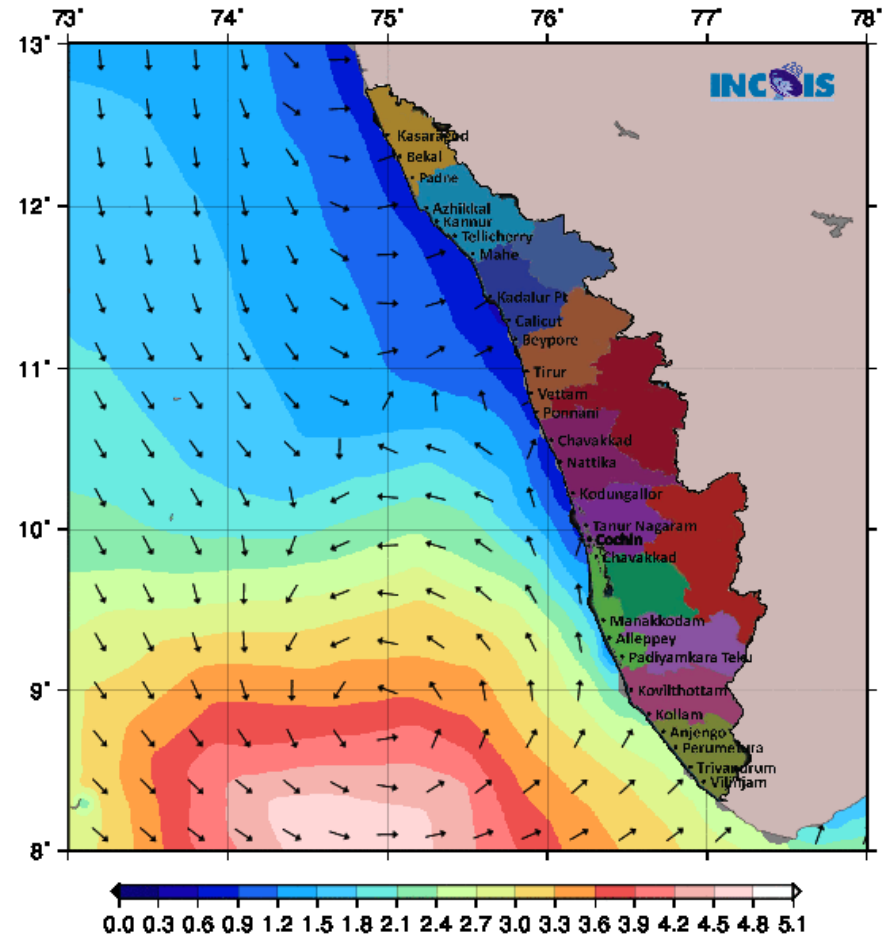
Forecast for 17.30 IST 12 OCT 2013



Arrows indicate direction of wave in degrees from North
Colour Scale indicate wave height in m

Significant Wave height (m) and Direction (°)

Forecast for 02.30 IST 01 Dec 2017



Arrows indicate direction of wave in degrees from North
Colour Scale indicate wave height in m

High swell/Kallakkadal events in the North Indian Ocean

➤Kurian et al., [2009] has reported a special case of coastal flooding by the long period (~ 15 s) swell waves on the southwest coast of India, named as *Kallakkadal*.

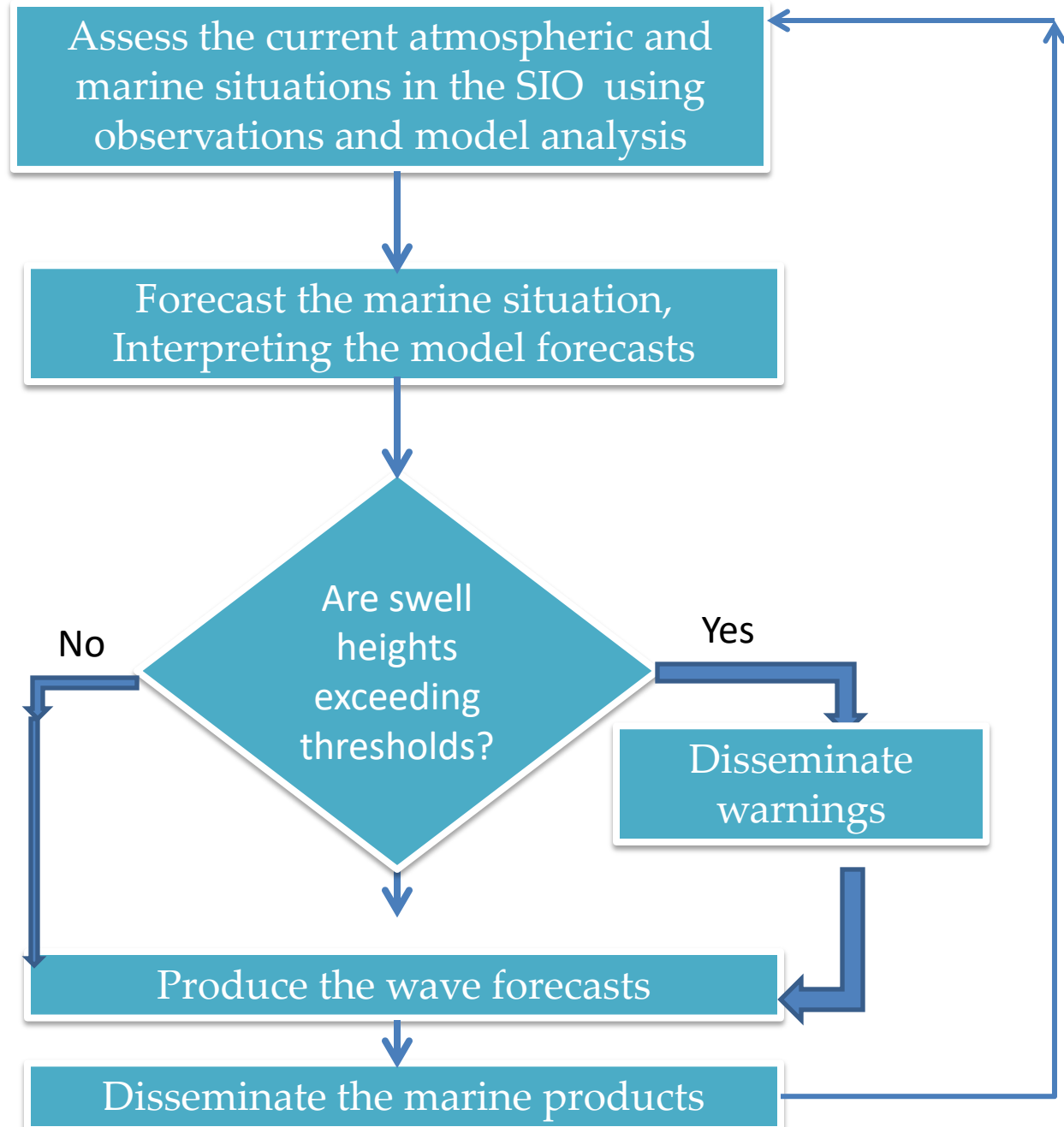
➤Kallakkadal is a flash flooding event without any precursors or any kind of local wind activity to give advance warning to the coastal population.

➤Long period swells (>14 s) that are having a moderate height (>0.4 m) and lasting for at least a half day (>12 hours)

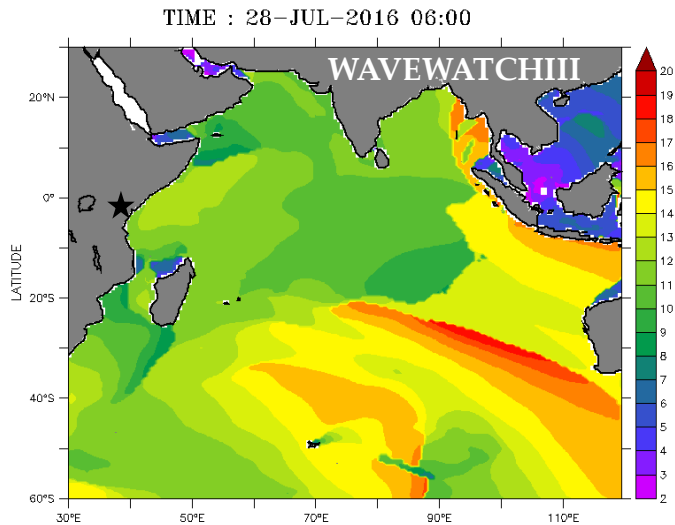


Flooding due to Wave Surge (Kallakadal) around January –February 2015 along Kollam coast, Kerala

Swell Forecast Process



KALLAKADAL EVENT DURING 30 JUL. - 03 AUG., 2016



High waves topping the embankment at old Digha, 03 Aug 2016, West Bengal

The Kollam District administration (DMD) :Wave surge was reported in coastal regions of Alappad Village of Karunagapally Taluk on on 1st and 2nd of August, 2016, due to The extra-tropical storm in the Southern Indian Ocean (27 Jul. 2016).

Various news paper Reports

Unexpected sea surge at Alappuzha coast, 27 fishing boats washed away, Indian Express, Published: 02nd August 2016



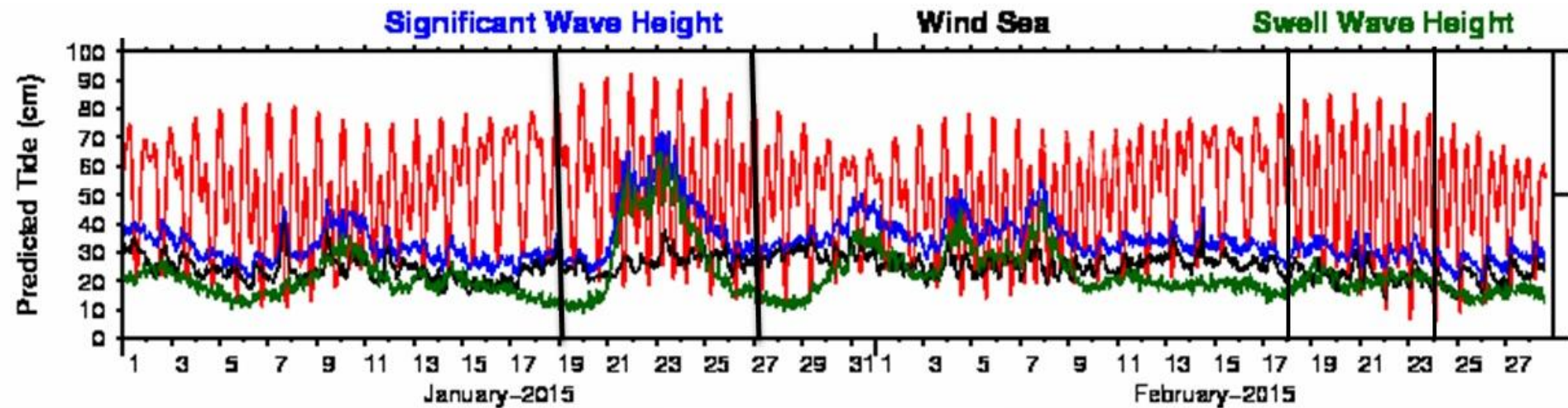
INCOIS high wave alert

➤INCOIS has issued timely wave surge alert for low lying coastal areas of Kerala from 30 Jul. 2016-03 Aug. 2016

➤A high wave, surge alert for the West Bengal Coast valid from 08:30 hours on 02-08-2016 to 23:30 hours of 03-08-2016 was issued by INCOIS.

- This information sent to all concerned disaster Management authorities and directly to fishermen via SMS. Total SMS sent (Tamilnadu, Orissa, Kerala, West Bengal, Gujarat, Maharastra, Lakshadweep) – 6965 ; Number of SMS sent to Kerala Fishermen – 340; Lakshadweep – 25

Wave Surge (Kallakadal) during January –February 2015



High Wave Bulletin:

EVENT-TYPE: Warning

ISSUE-DATE: 22-01-2015

REGION: Kerala

MESSAGE: High swell waves in the range of 2.0 - 2.3 meters are forecasted during 17:30 hours on 22-01-2015 to 2330 hours of 24-01-2015 along the Kerala coast between Vizhinjam to Kasargod.

MESSAGE: Because of the combined effect of Spring tide and high waves, waves may be surged in to low lying coastal areas intermittently.

MESSAGE: Fishermen are advised to be cautious while venturing into the sea. Tourists also advised cautious while venturing into sea.



Flooding along Kollam coast, Kerala

Oil Spill

INCOIS OIL SPILL ADVISORY SYSTEM



Release of liquid petroleum hydrocarbon into the marine environment is called Oil Spill



Carbon	83-87 wt %
Hydrogen	10-14 wt %
Nitrogen	0.1-2 wt %
Oxygen	0.1-1.5 wt %
Sulfur	0.5-6 wt %
Metals	< 0.1 wt %

**Elemental composition
of petroleum crude oil**

Based on the quantity
(ICG Norms)

NEED FOR AN OIL SPILL TRAJECTORY PREDICTION SYSTEM

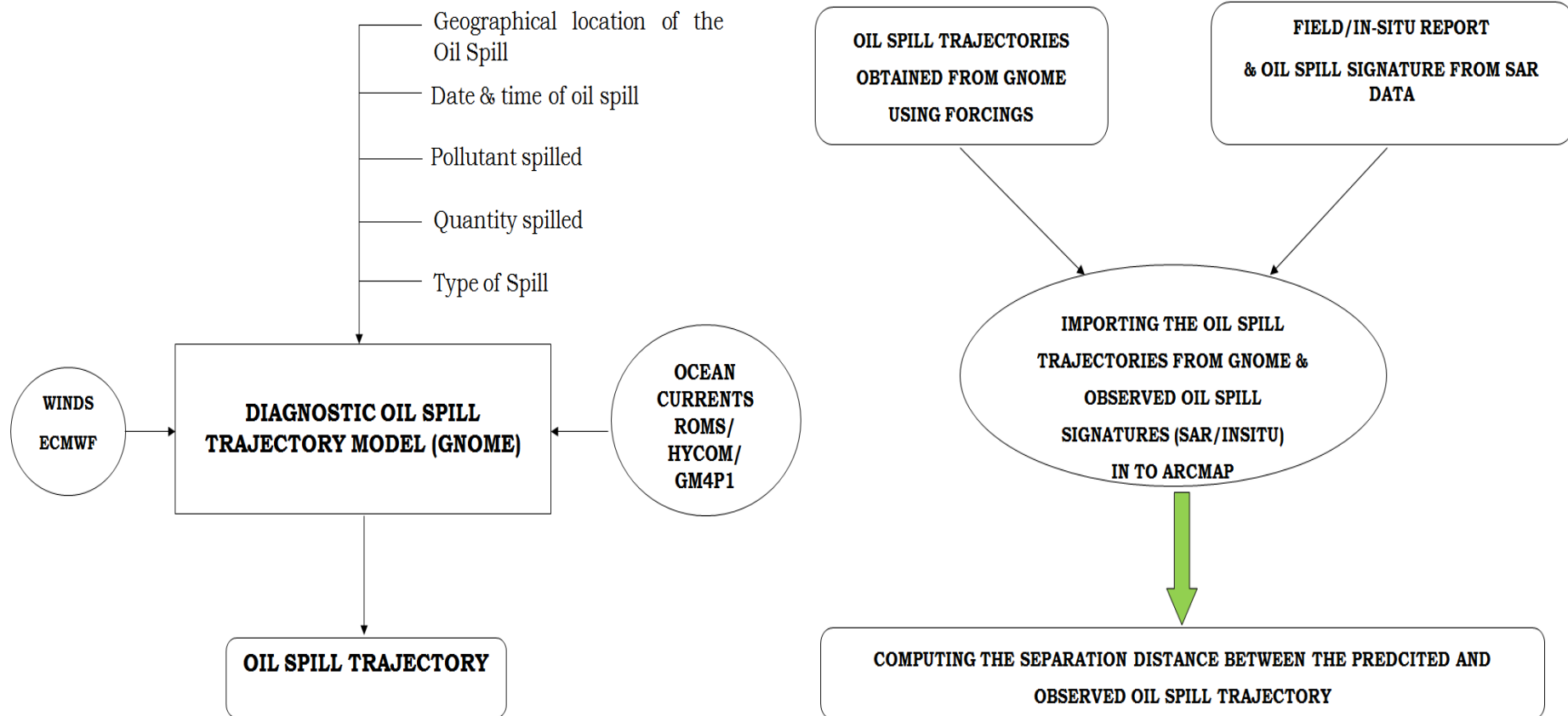
In order to prevent the impact of oil spills on the marine environment an oil spill trajectory prediction system is required. In the event of oil spill, the direction and movement of the oil will be predicted in advance in our system and will be disseminated to the Regulatory Authority. The clean up and control measures will be planned and carried out accordingly.

Tier -1 < 700 Tons

Tier -2 <10000 Tons

Tier - 3 > 10000 Tons

OIL SPILL TRAJECTORY PREDICTION & VALIDATION SET UP AT INCOIS

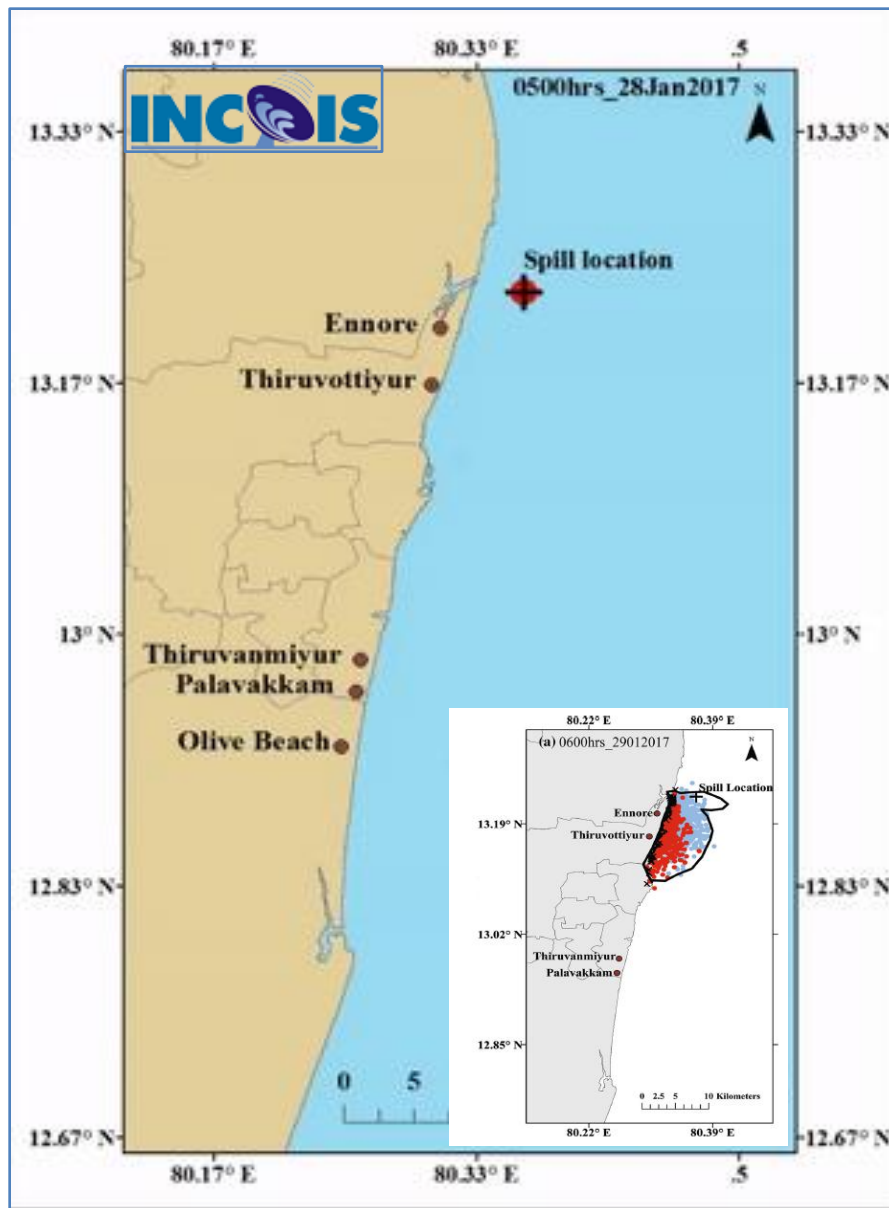


CATEGORY OF ACTIVITIES

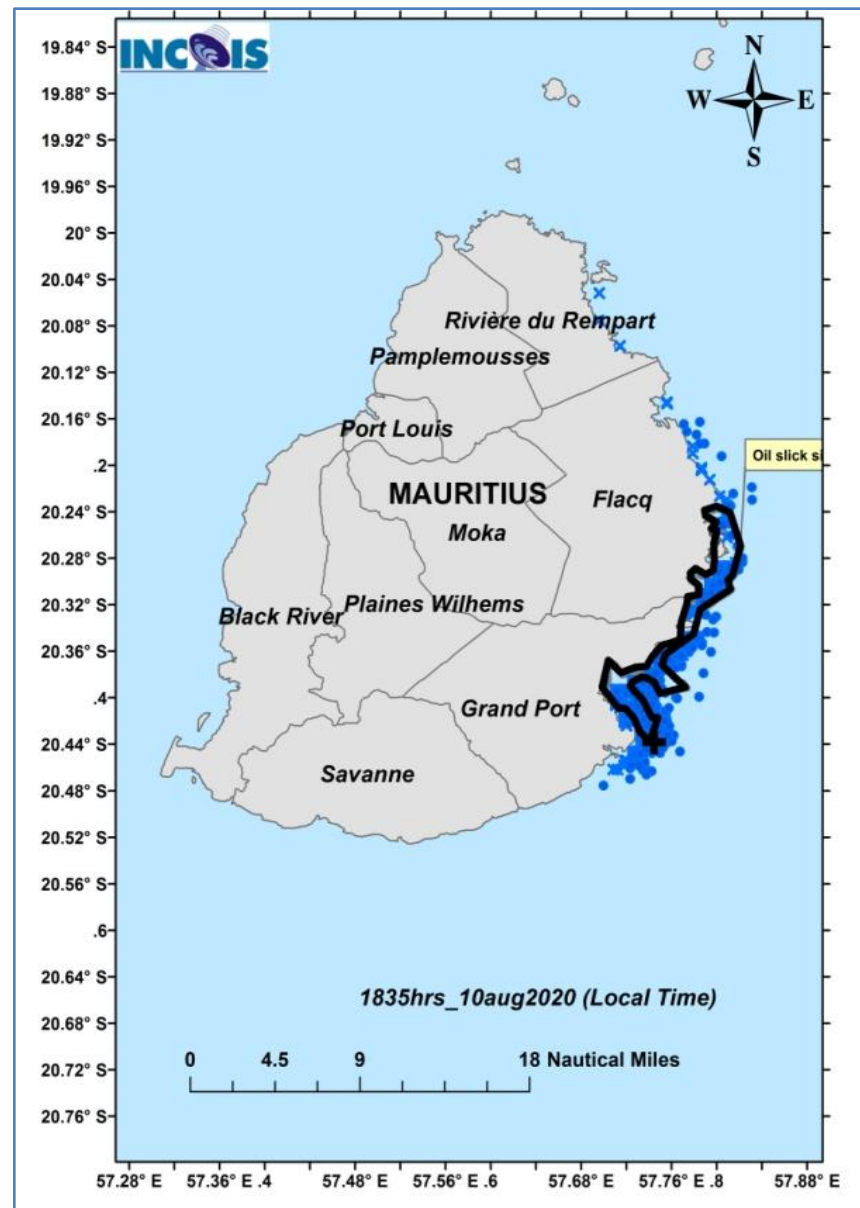
- **Advisory Tasks – Issuing oil drift patterns (Real, Hypothetical oil spills, Marine Mock drills and POLREX)**
- **Operational tasks – Maintaining and upgrading the system**
- **Research Tasks – Modelling studies /SAR data process**
- **Consultancy works – Impact of oil spill in port development projects**

INCOIS AID IN OIL SPILL RESPONSE EVENTS (INDIA AND NEIGHBOUR COUNTRIES)

ENNORE OIL SPILL (PREDICTION & VALIDATION)



MAURITIUS OIL SPILL (PREDICTION & VALIDATION)



Black polygon in both frames denotes the oil slick signature from SAR data

User comments on Ocean State Forecast from INCOIS



We are blessed with the helpline facility for delivering OSF information and this facility plays a significant role in our lives. My team can now proceed for any fishing activity without any fear

Antony, 28, Fisherman, Kolachal, TN.



PMSSS on behalf of the target communities expresses its sincere appreciation and thanks to INCOIS for its valuable, timely advise, support and guidance during emergency situations- **PMSSS, TN.**



The OSF data/images are very accurate and useful which keeps us updated during sailing. The OSF reports are very important for our passenger vessels sailing always in low pressure areas like Andaman sea- **Master MV Swarajdweep, Shipping Corporation of India.**



It has been observed during the recent operations that the forecast provided by INCOIS has closely matched with that of the actuals and has been well appreciated by the operation co-coordinator- **Commander Mangal Kakkad, Navy.**



"The technology developed by the Ocean State Forecast Division of INCOIS has however changed this scenario and has made the fishermen confident of the wave heights at different distances from the shore line. They now go with great joy and courage into the Sea and they also now know where approximately the fish shoals are. Thus, INCOIS has helped to transform the lives and livelihoods of small scale fisher families. This has let to the birth and spread of a science and technology for artisanal fisheries movement". **Prof. Dr. M. S. Swaminathan, MSSRF, Chennai, Current Science, 105 (2), pp. 175-181, 2014.**

Thank You